

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

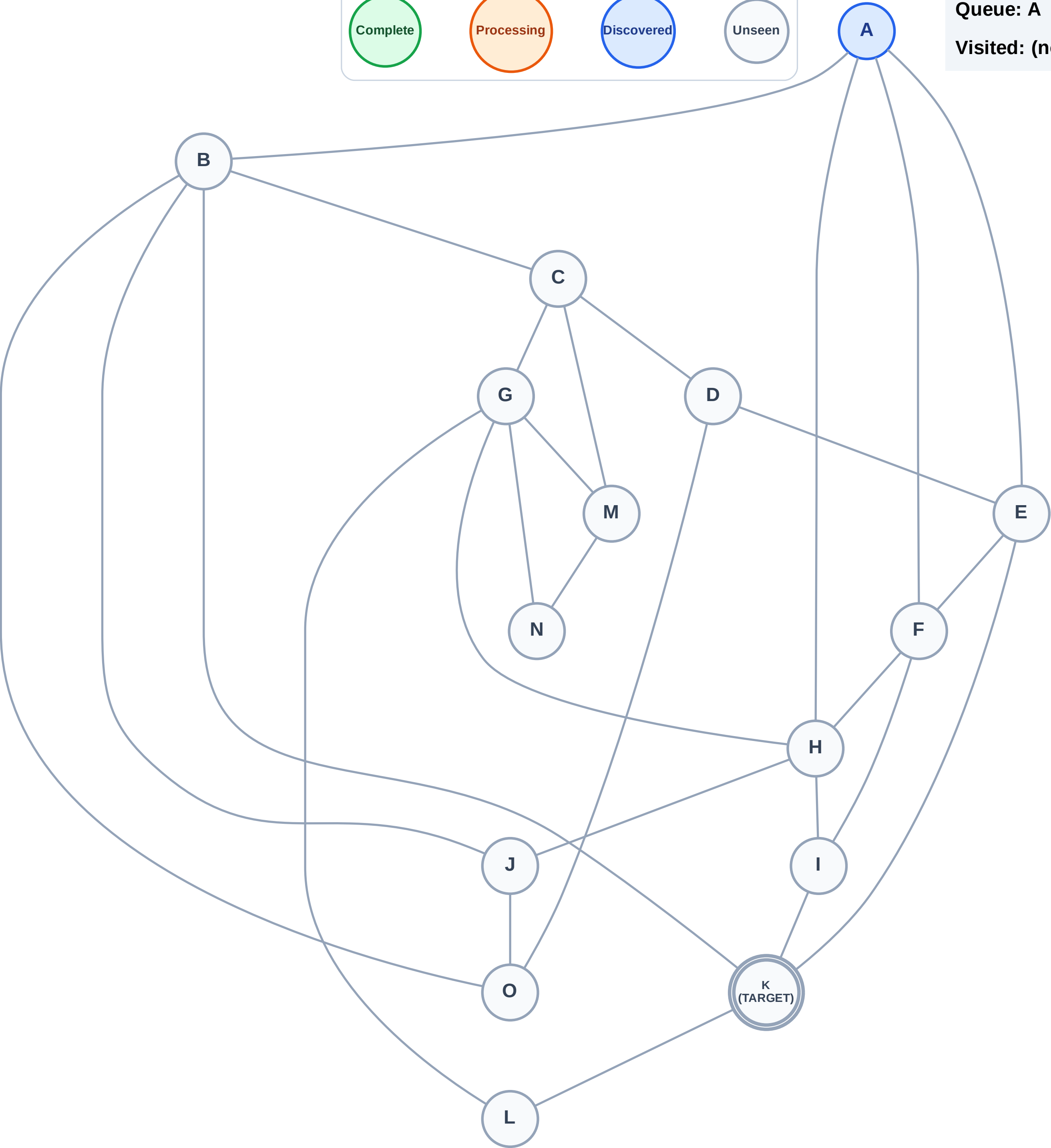
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

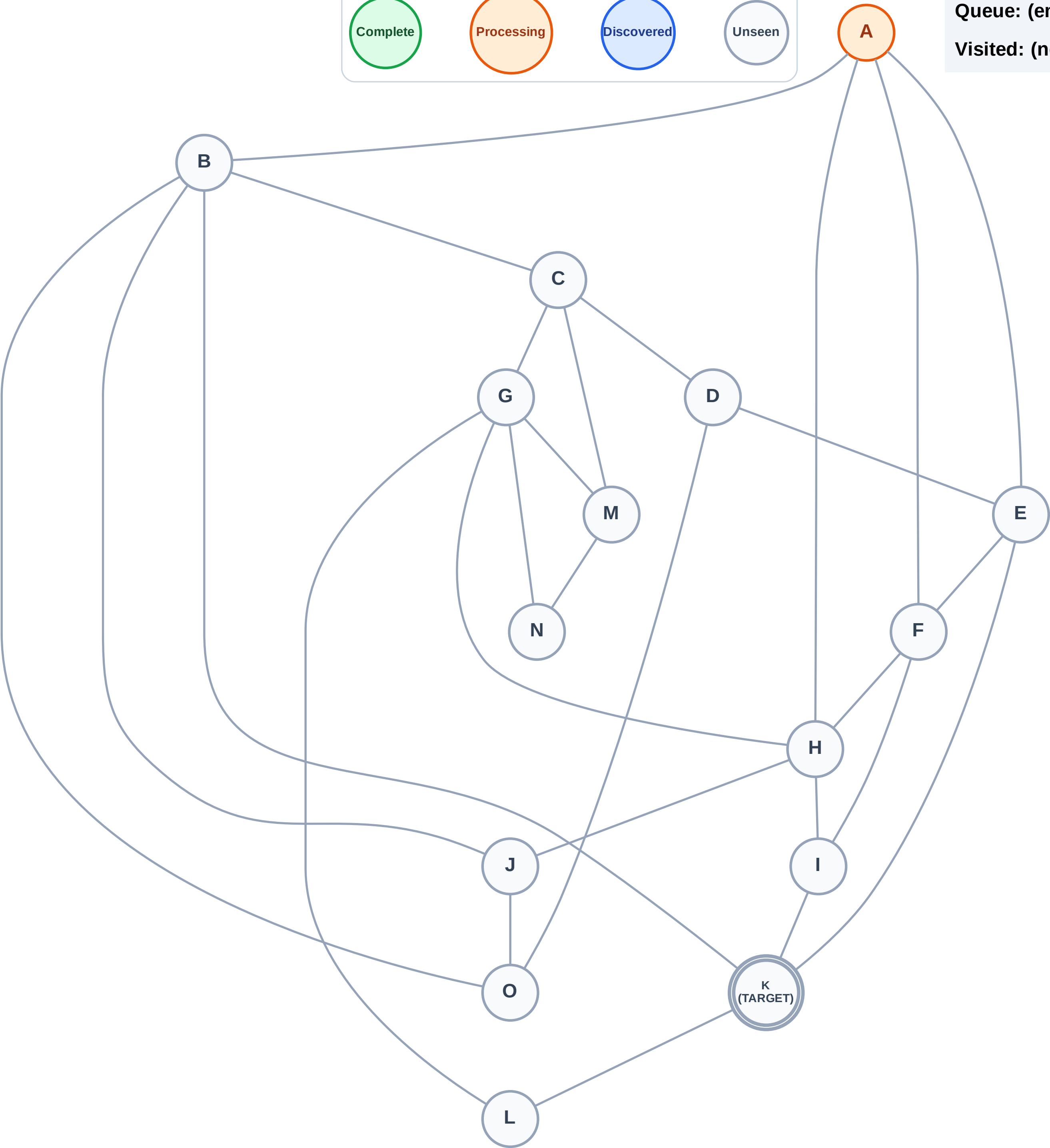
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, E, F, H from A

Legend



Complete



Processing

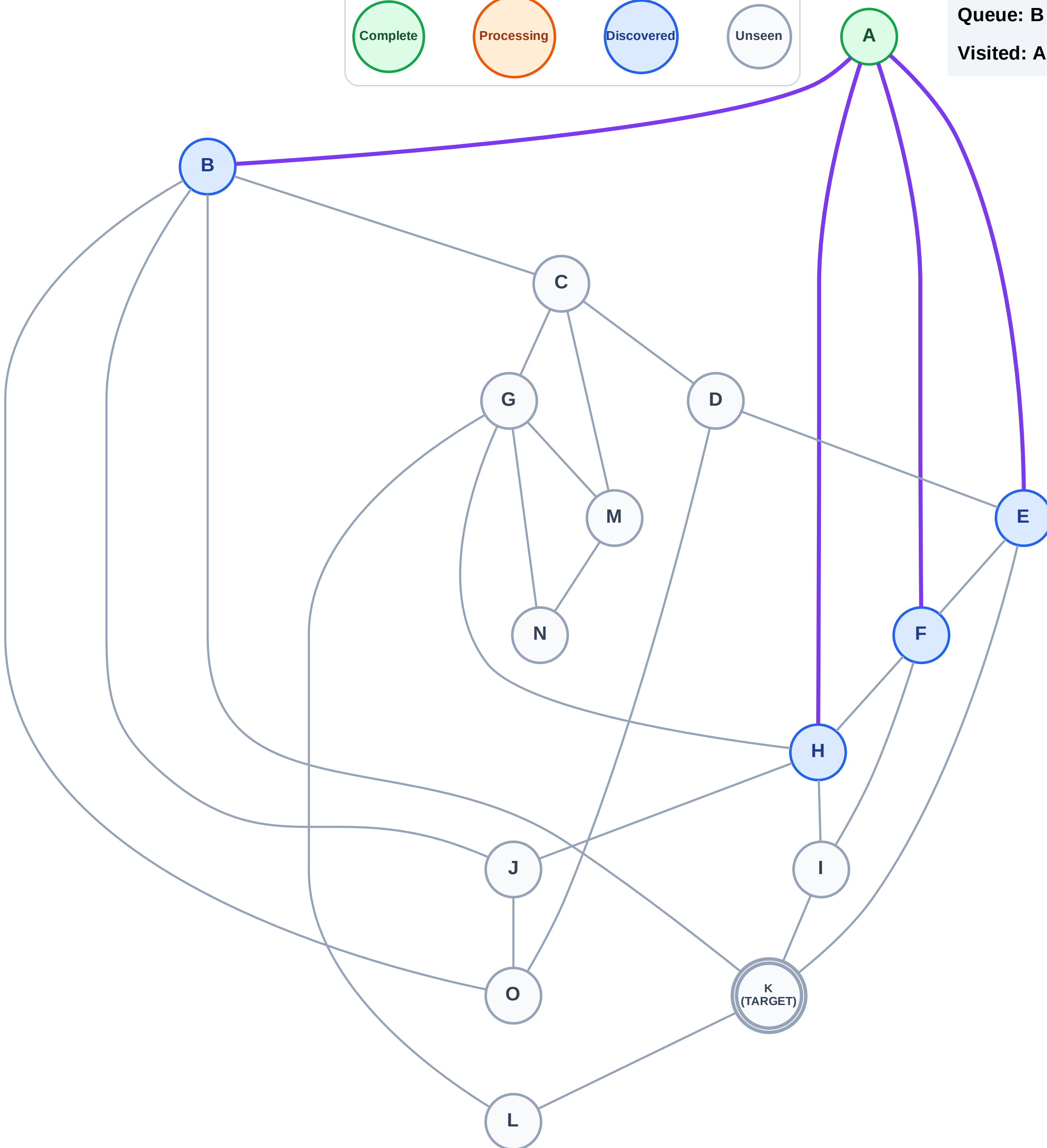


Discovered



Unseen

Queue: B → E → F → H
Visited: A



BFS Traversal

Step 4: Process node B

Legend

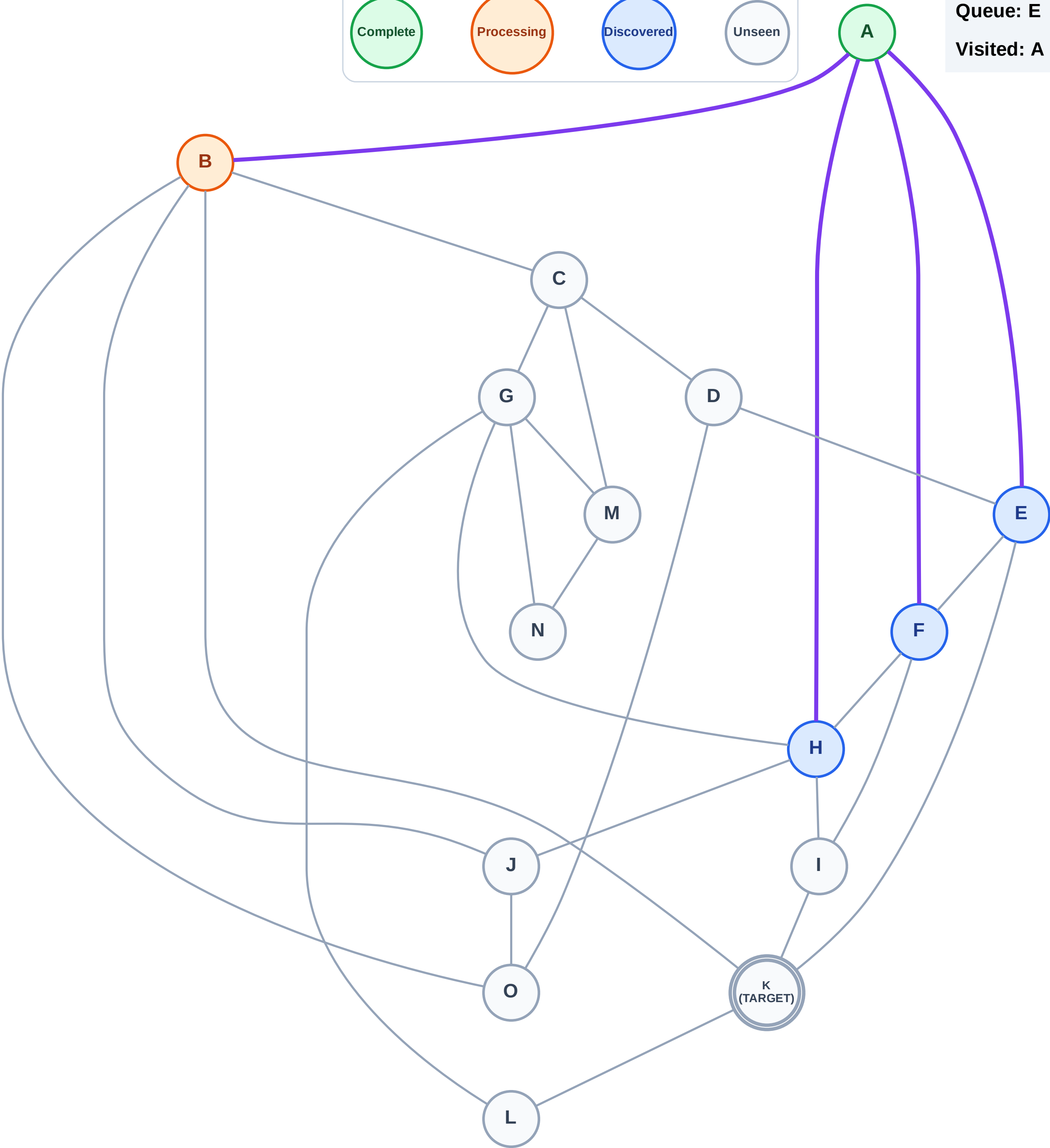
Complete

Processing

Discovered

Unseen

Queue: E → F → H
Visited: A



BFS Traversal

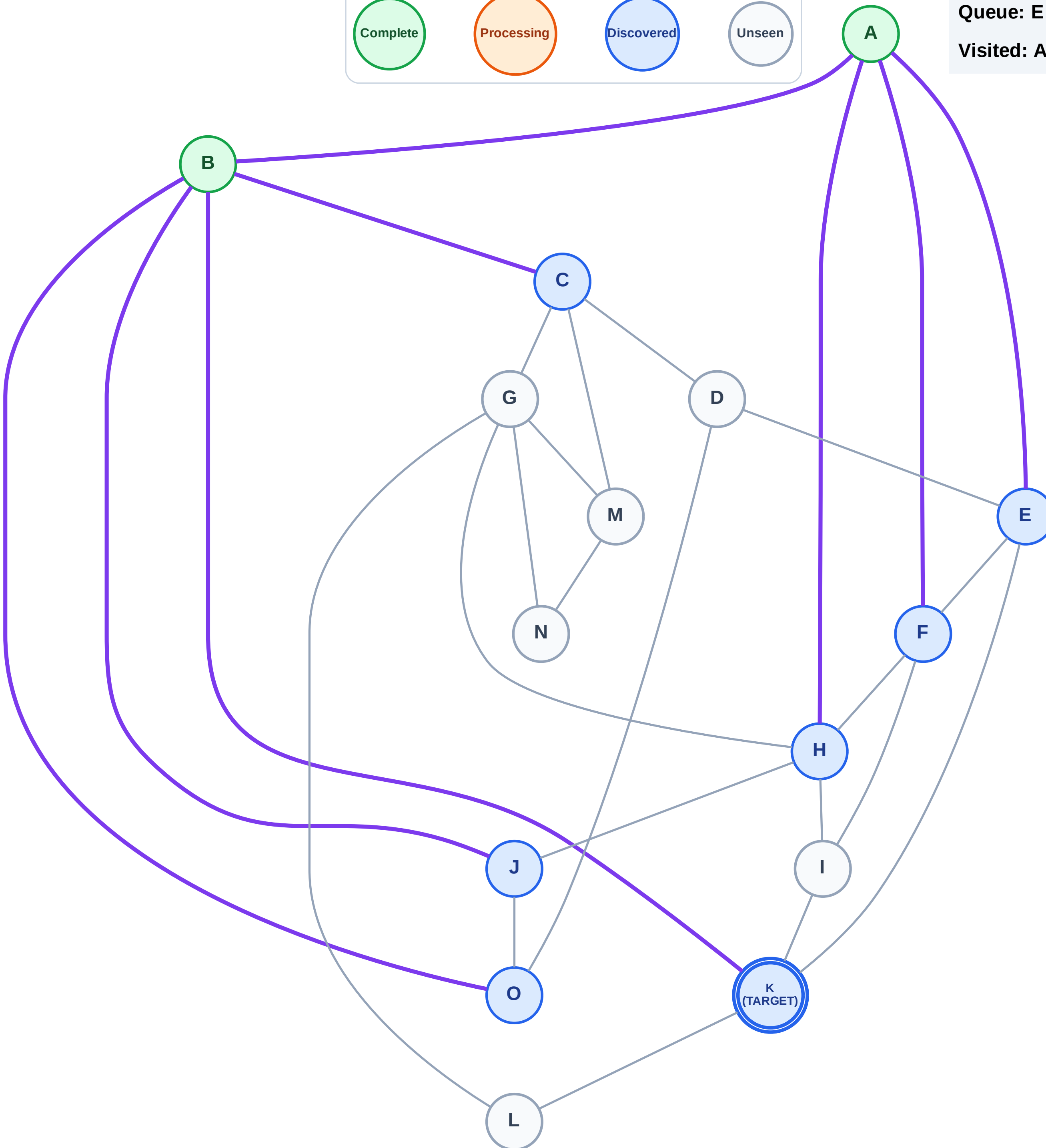
Step 5: Discover C, J, K, O from B

Legend



Queue: E → F → H → C → J → K → O

Visited: A, B



BFS Traversal

Step 6: Process node E

Legend

Complete

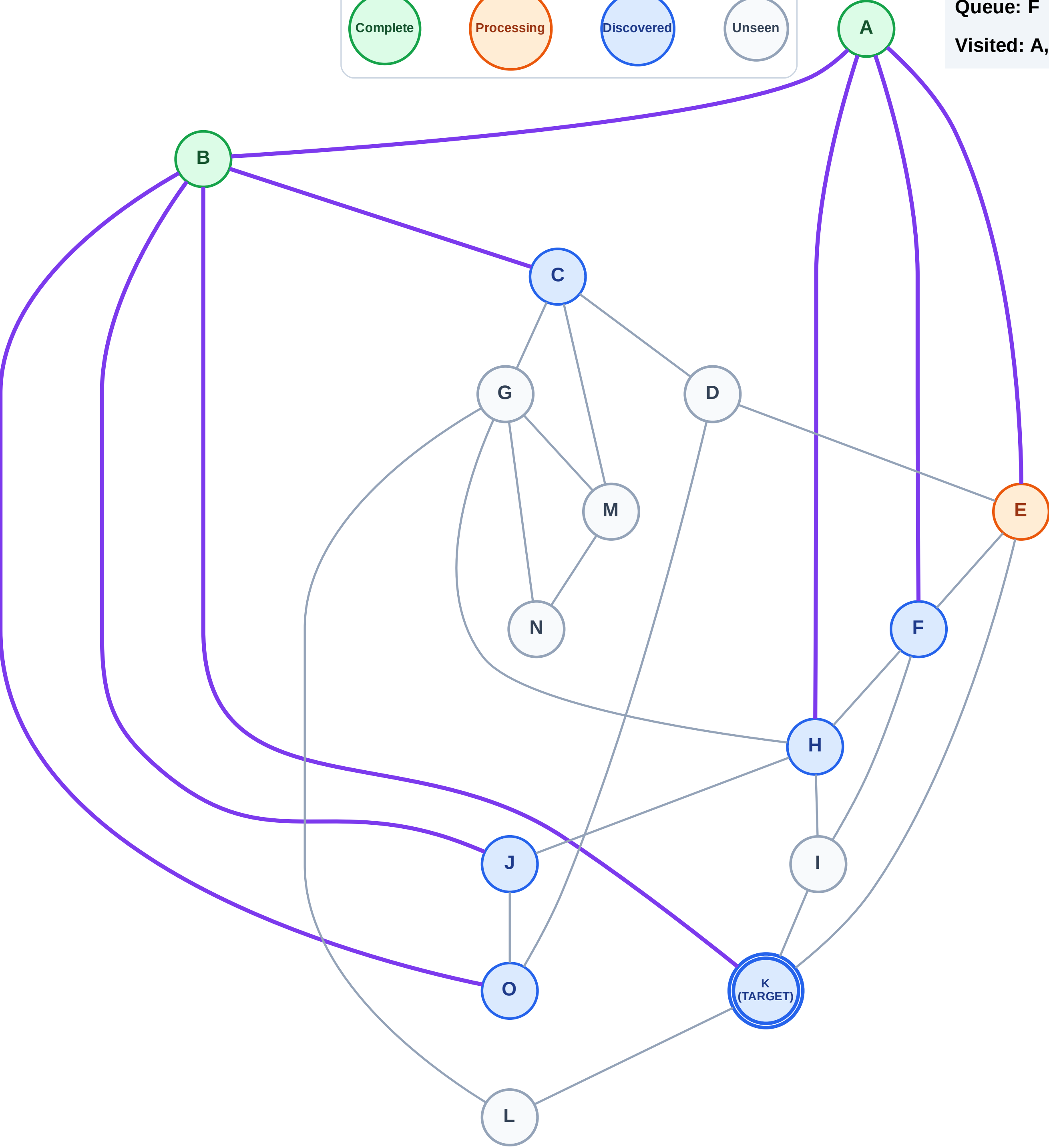
Processing

Discovered

Unseen

Queue: F → H → C → J → K → O

Visited: A, B



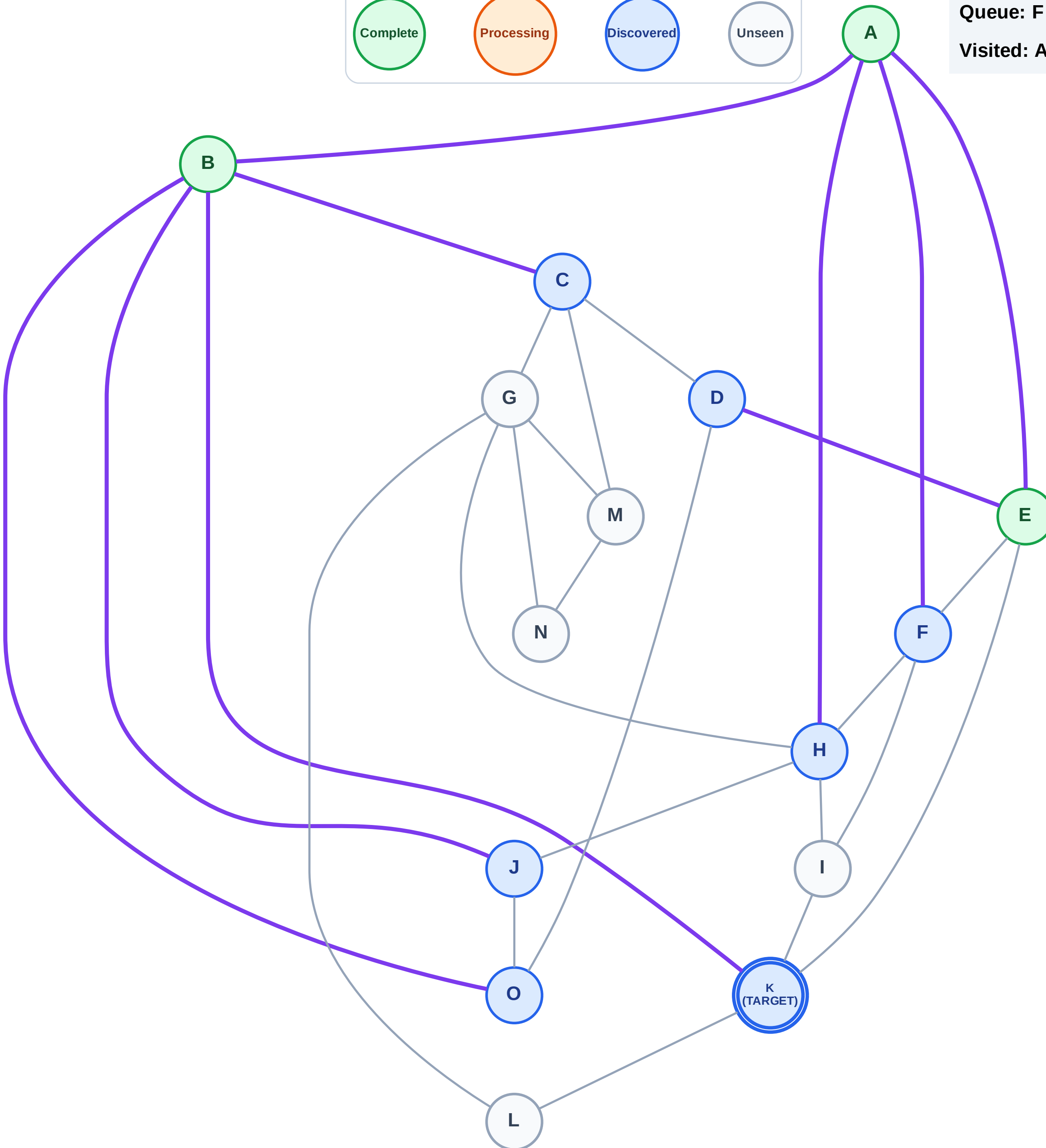
BFS Traversal

Step 7: Discover D from E

Legend



Queue: F → H → C → J → K → O → D
Visited: A, B, E



BFS Traversal

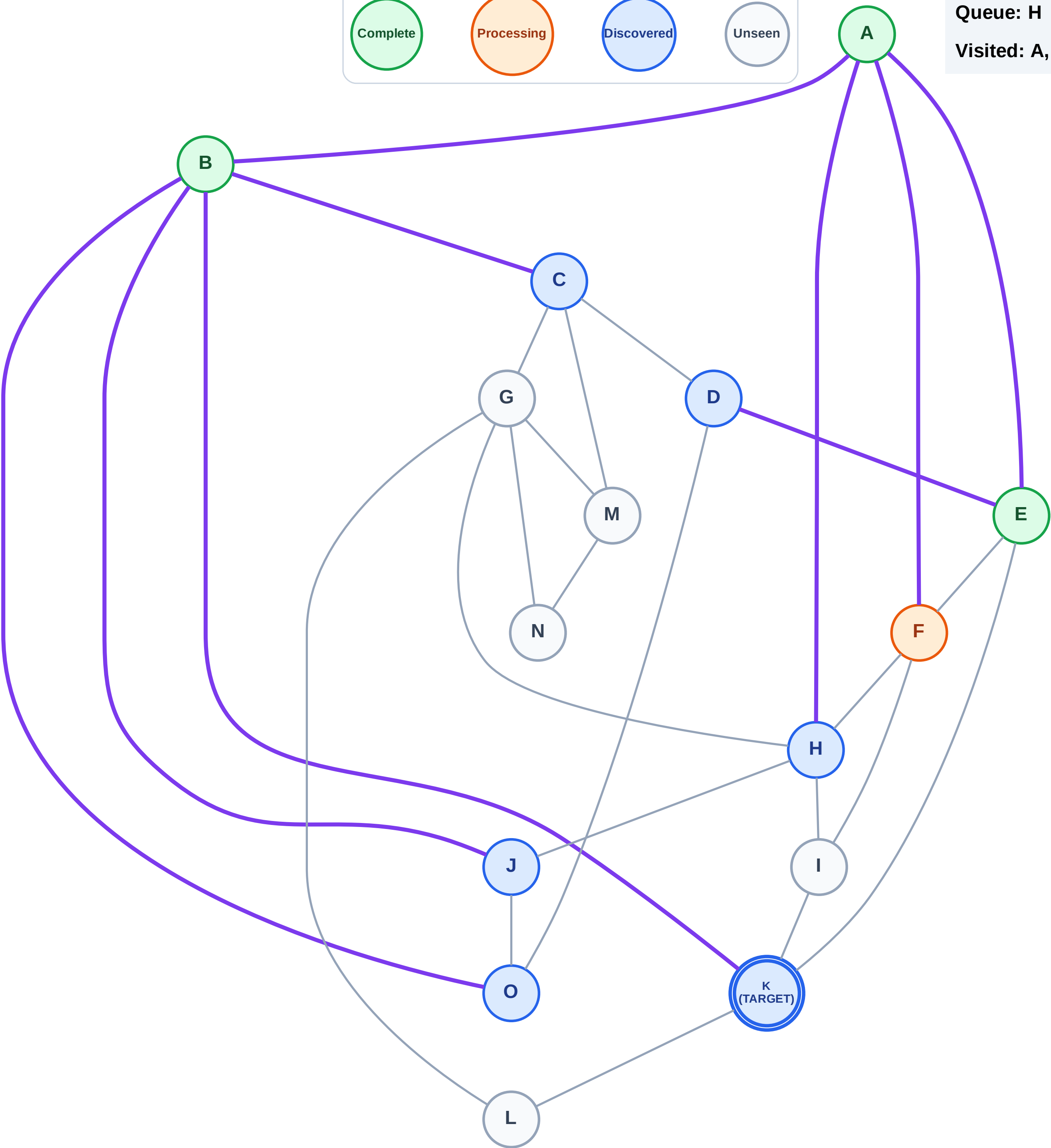
Step 8: Process node F

Legend



Queue: H → C → J → K → O → D

Visited: A, B, E



BFS Traversal

Step 9: Discover I from F

Legend

Complete

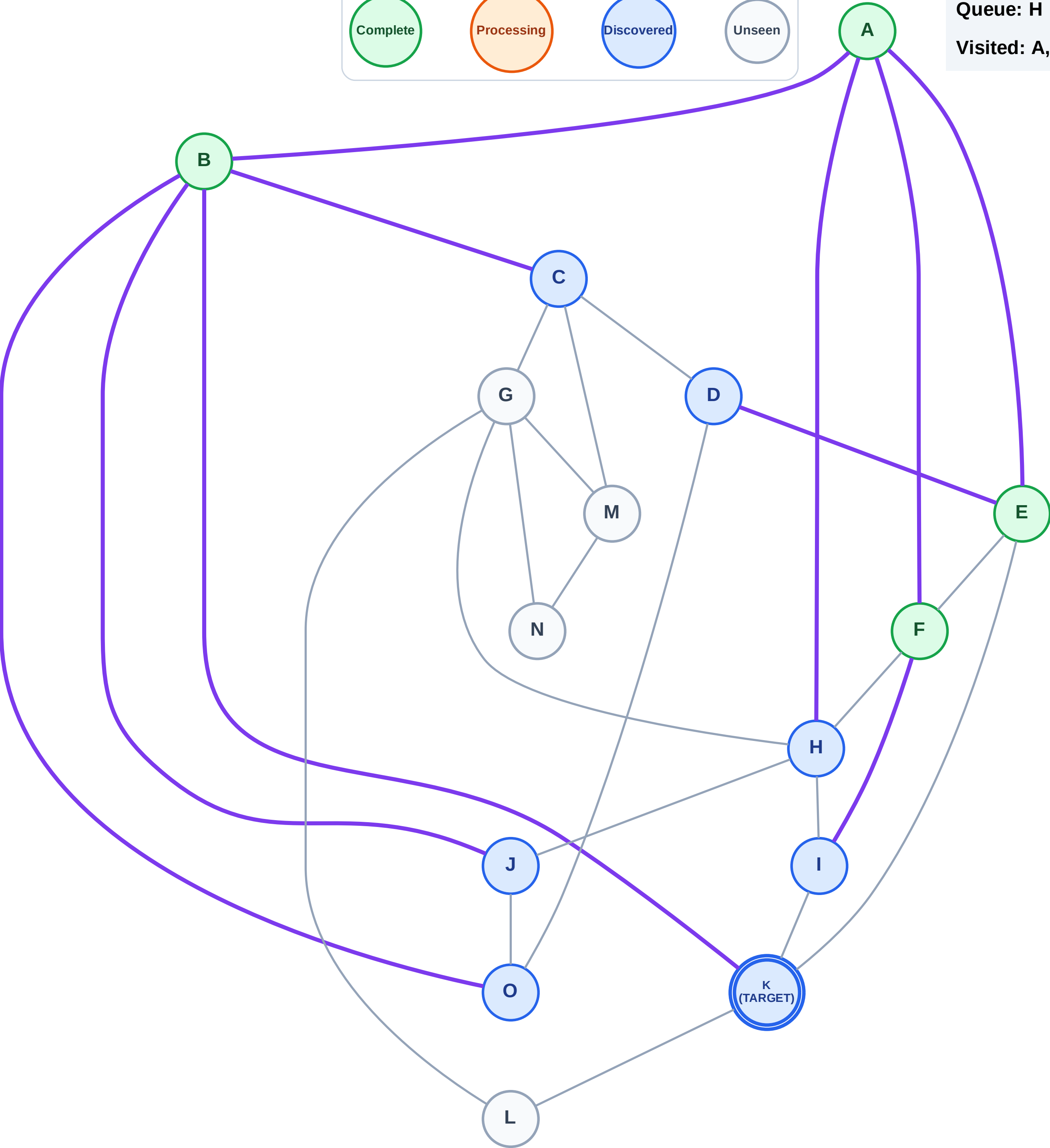
Processing

Discovered

Unseen

Queue: H → C → J → K → O → D → I

Visited: A, B, E, F



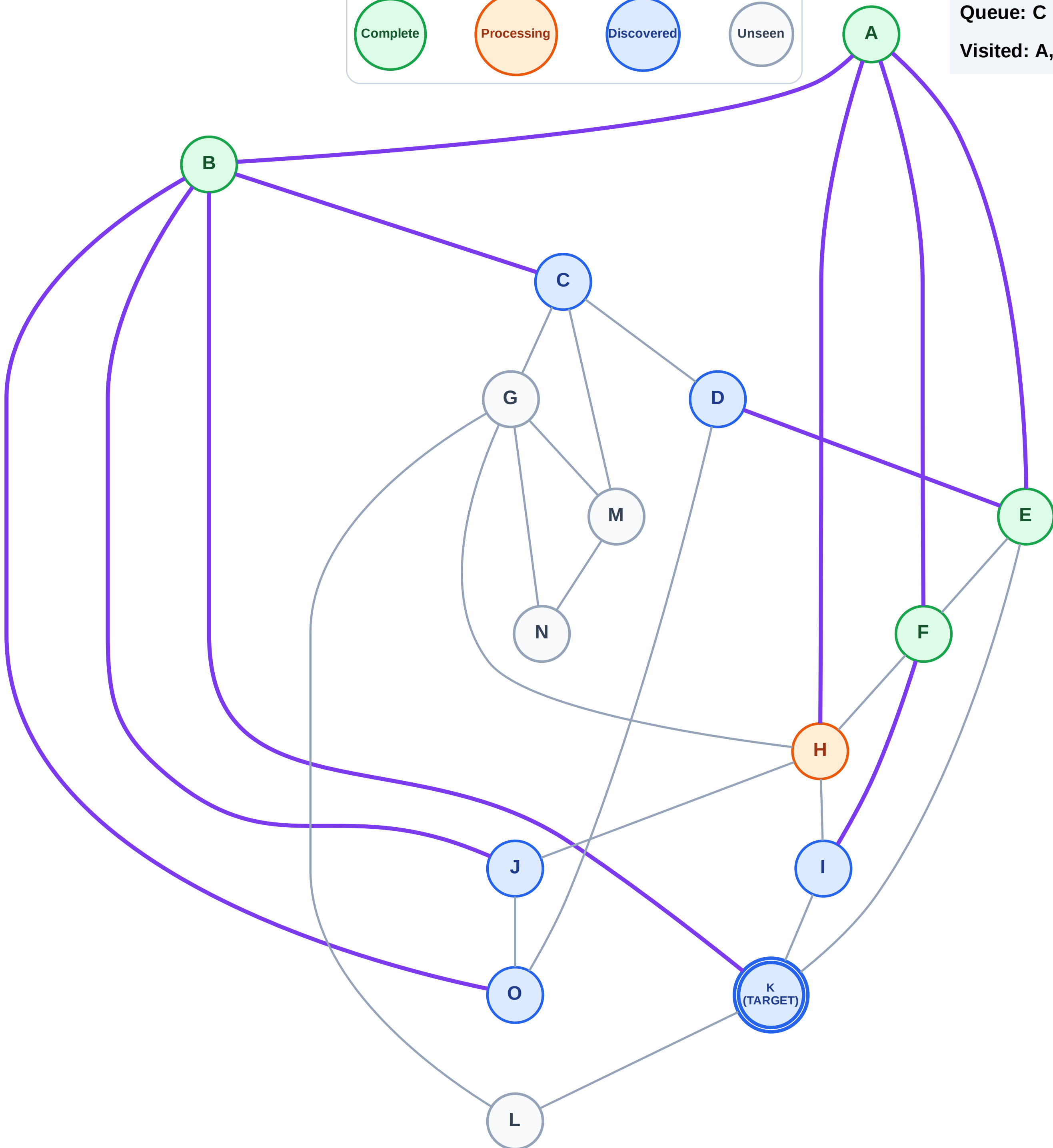
BFS Traversal

Step 10: Process node H

Legend



Queue: C → J → K → O → D → I
Visited: A, B, E, F



BFS Traversal

Step 11: Discover G from H

Legend



Complete



Processing

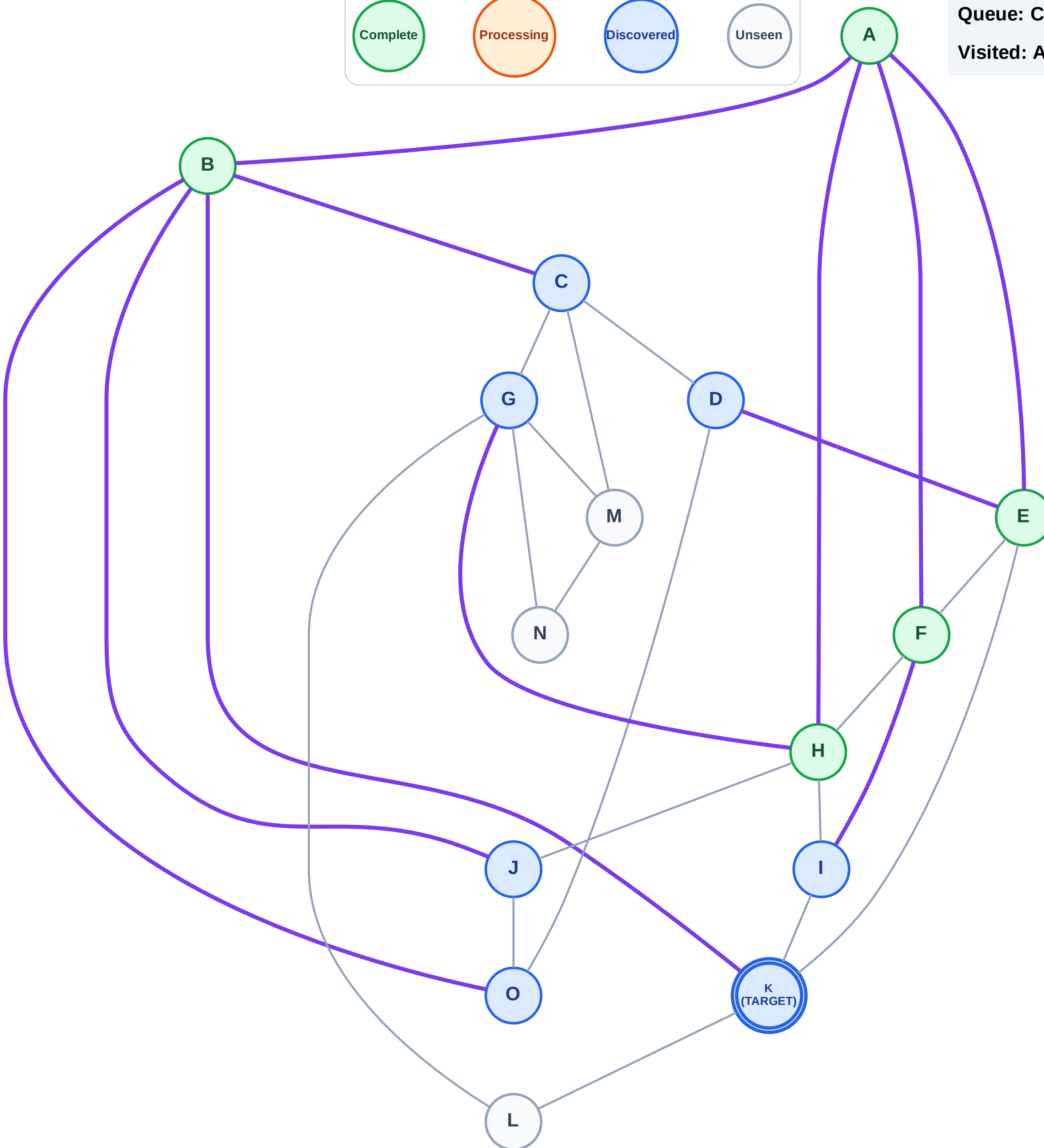


Discovered



Unseen

Queue: C → J → K → O → D → I → G
Visited: A, B, E, F, H



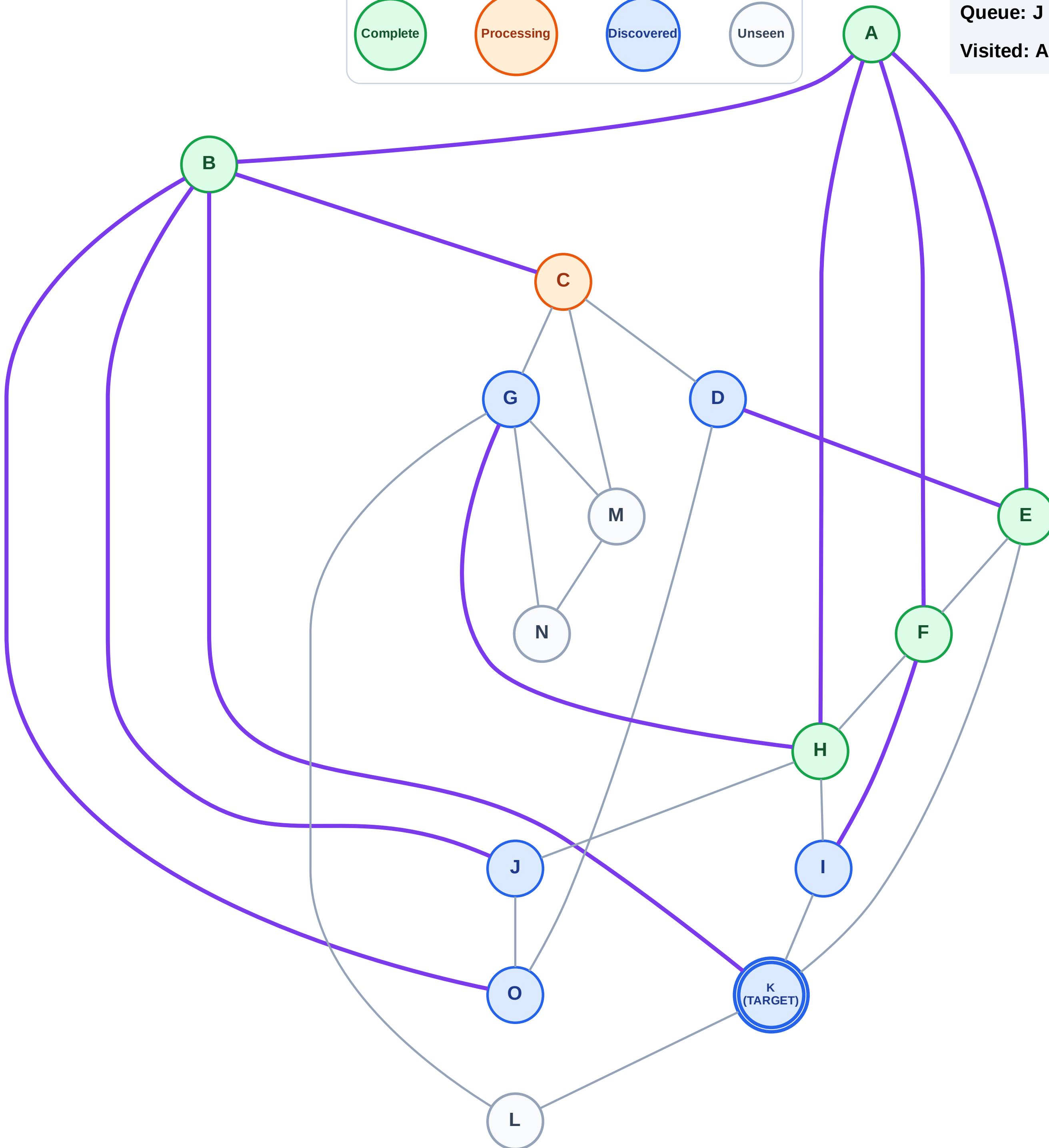
BFS Traversal

Step 12: Process node C

Legend



Queue: J → K → O → D → I → G
Visited: A, B, E, F, H



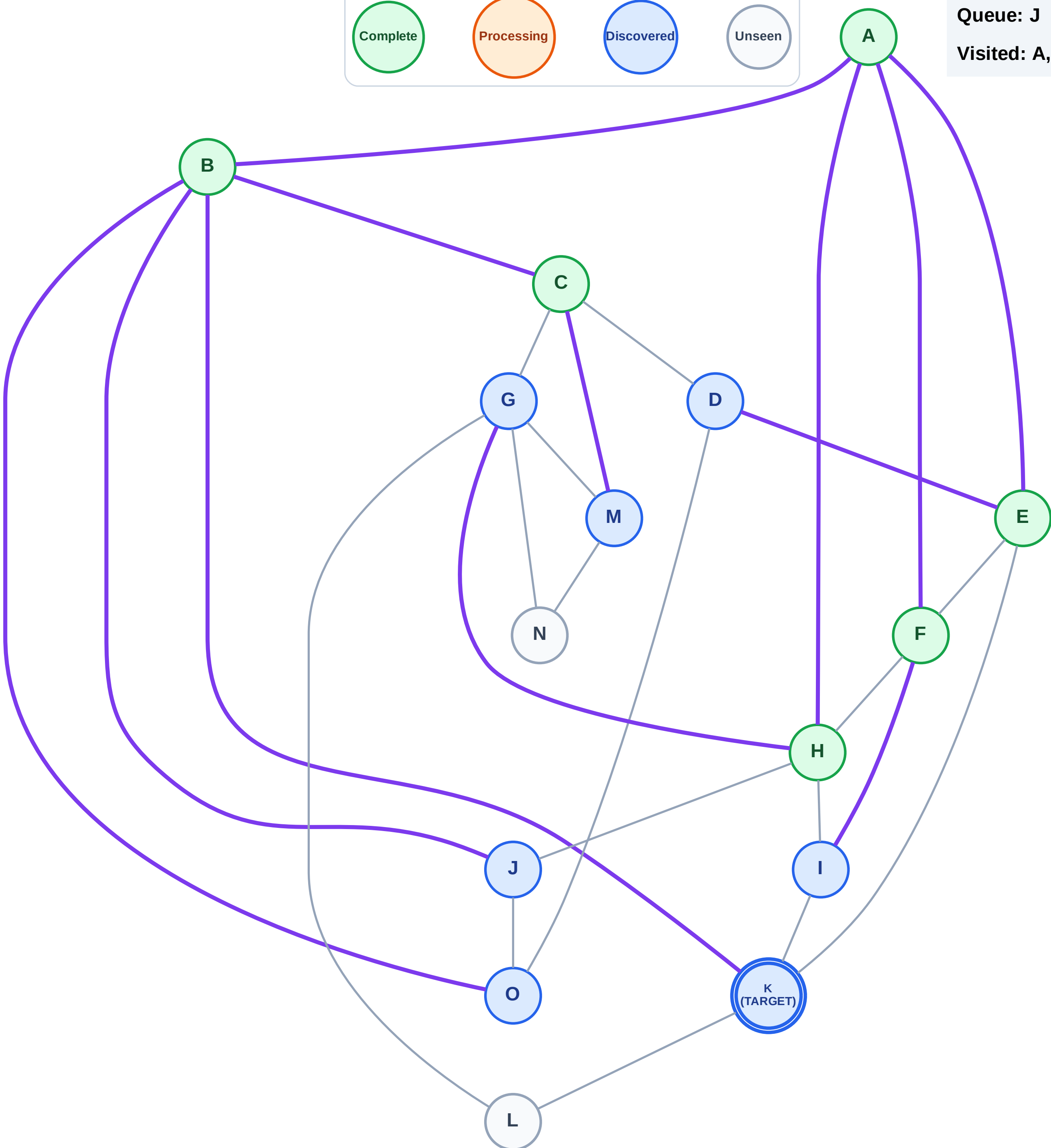
Step 13: Discover M from C

Legend



Complete Processing Discovered Unseen

Visited: A, B, C, E, F, H



BFS Traversal

Step 14: Process node J

Legend



Complete



Processing



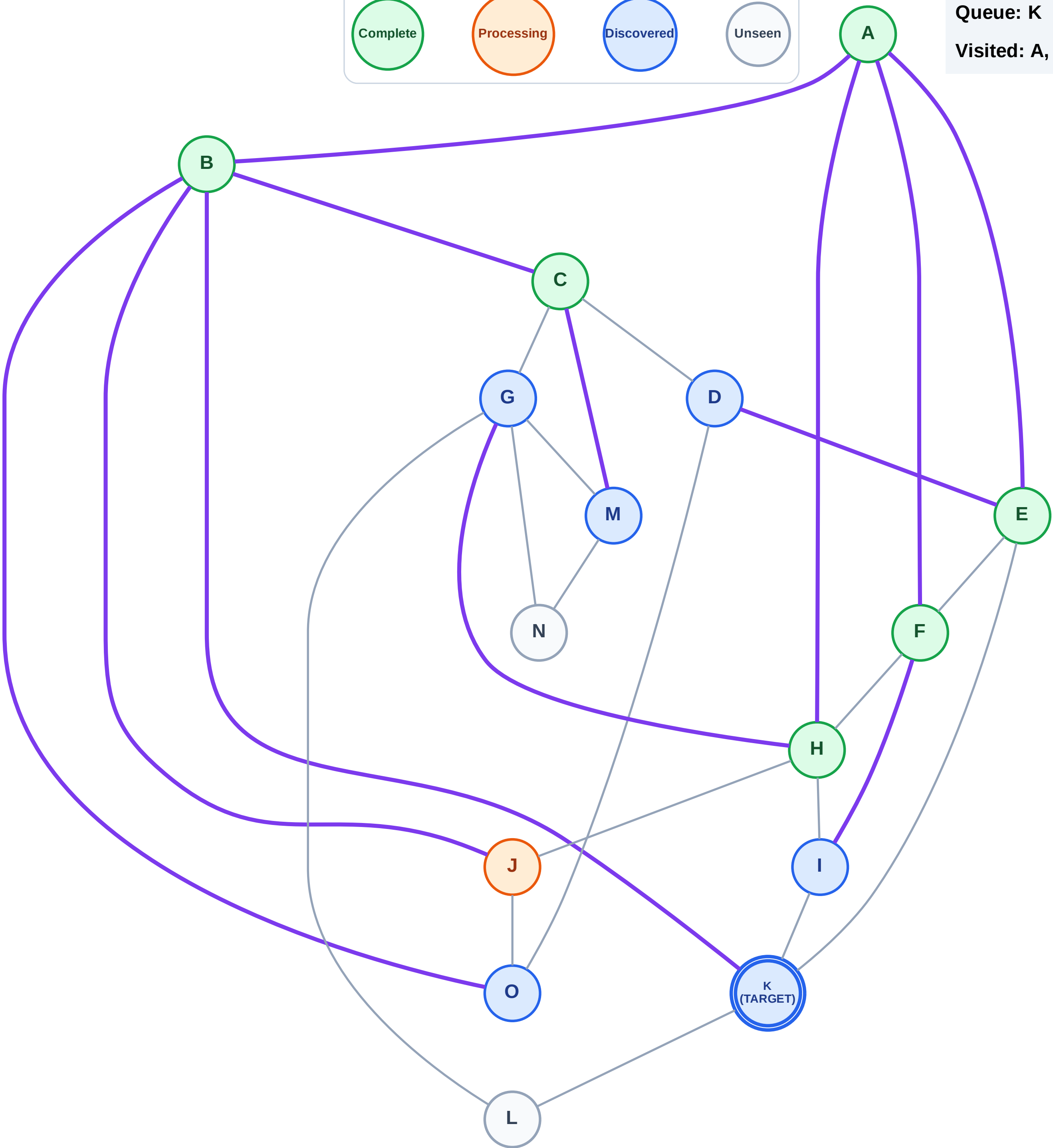
Discovered



Unseen

Queue: K → O → D → I → G → M

Visited: A, B, C, E, F, H



BFS Traversal

Step 15: No new neighbors from J

Legend



Complete



Processing

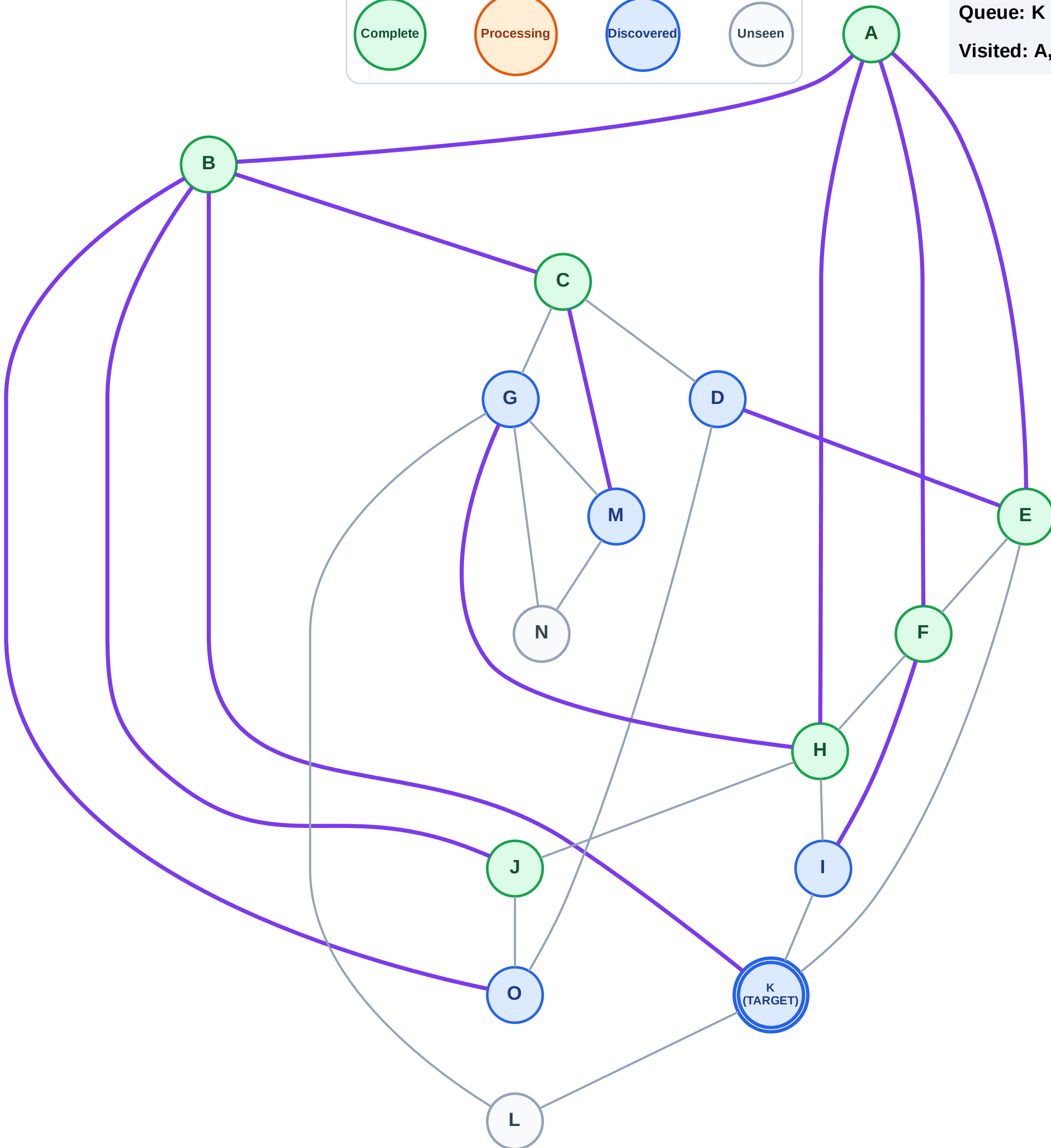


Discovered



Unseen

Queue: K → O → D → I → G → M
Visited: A, B, C, E, F, H, J



BFS Traversal

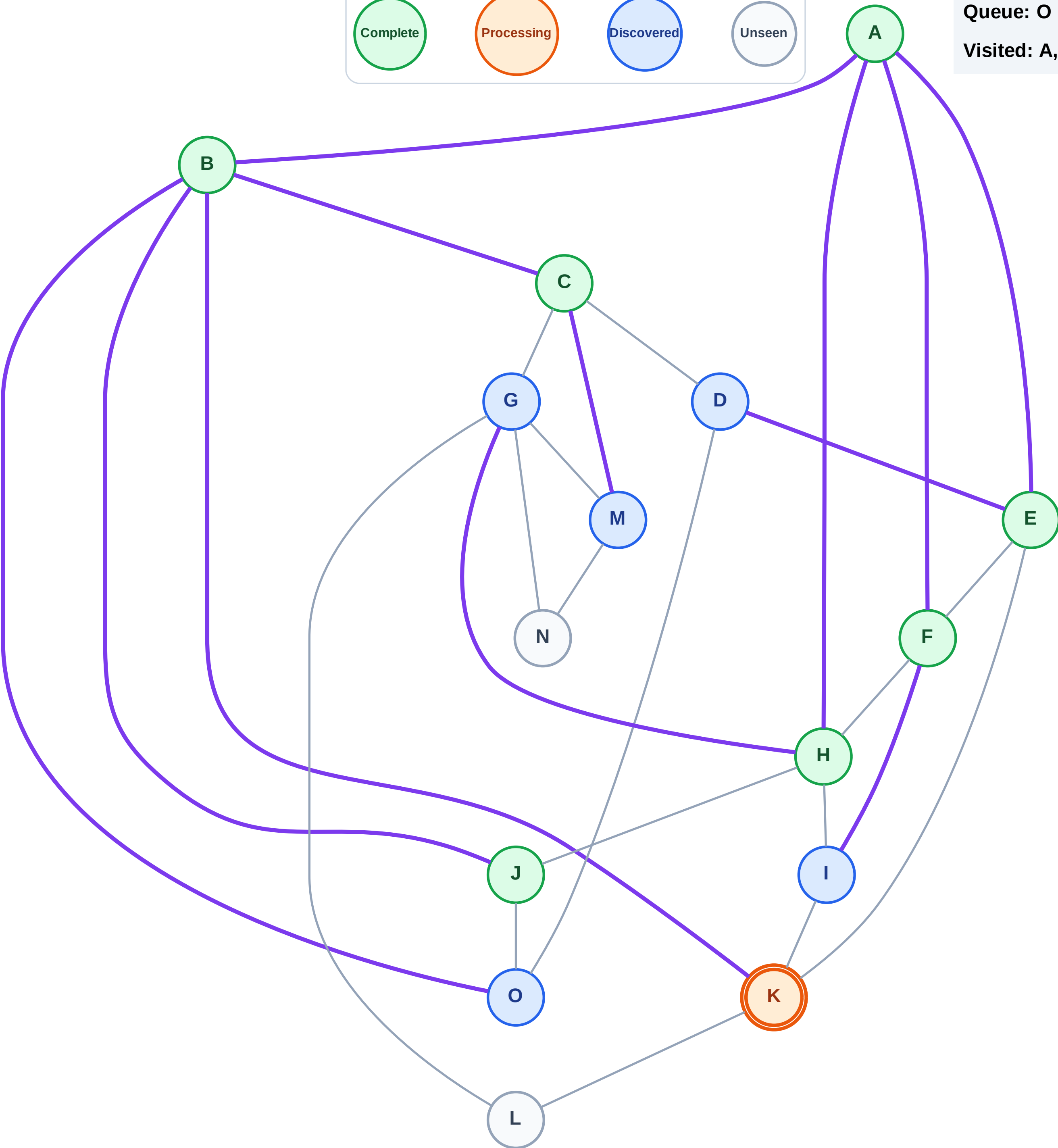
Step 16: Process node K

Legend



Queue: O → D → I → G → M

Visited: A, B, C, E, F, H, J



BFS Traversal

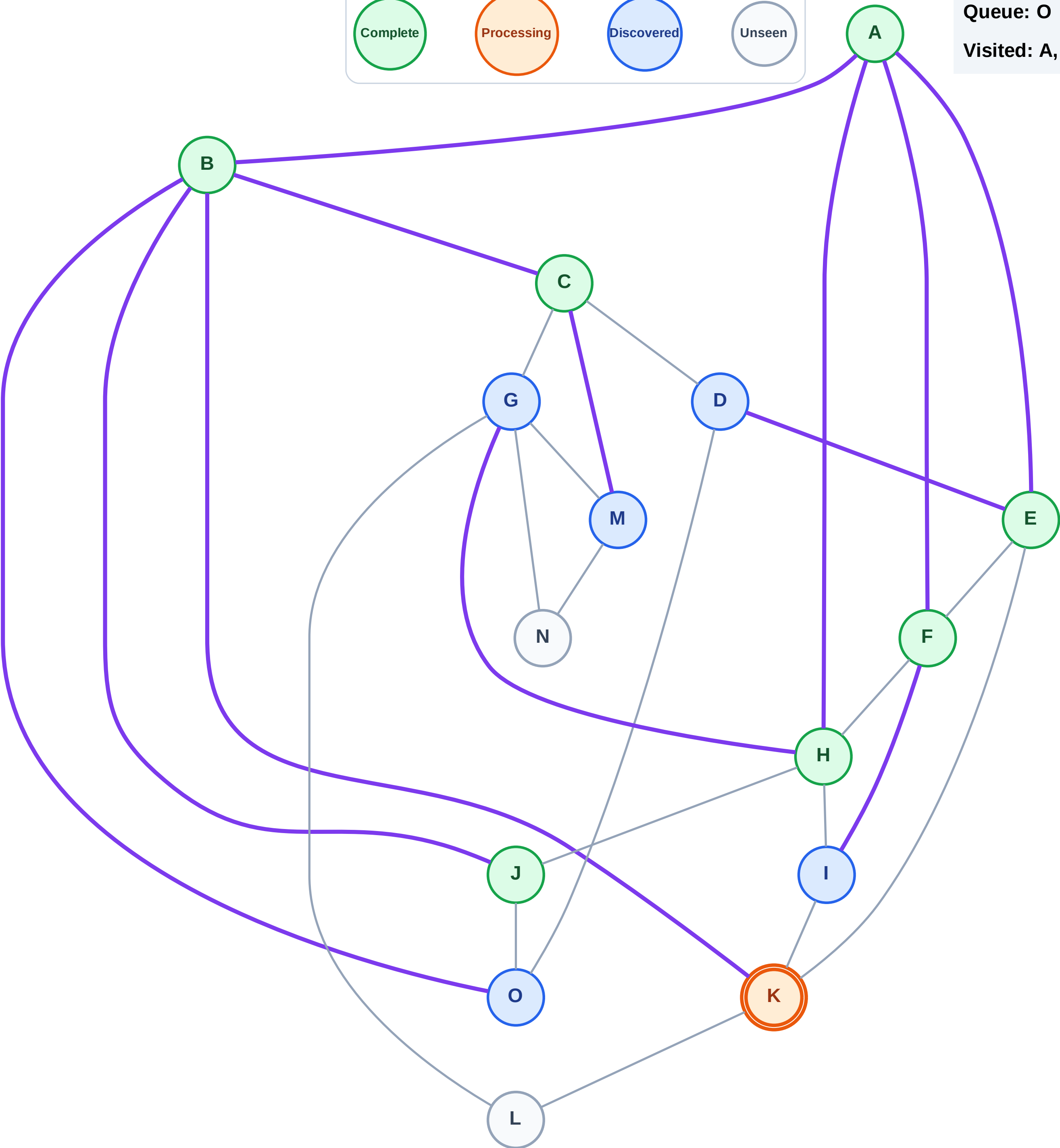
Step 17: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Queue: O → D → I → G → M

Visited: A, B, C, E, F, H, J, K



Step 1: Start at A (Seeking target node K)

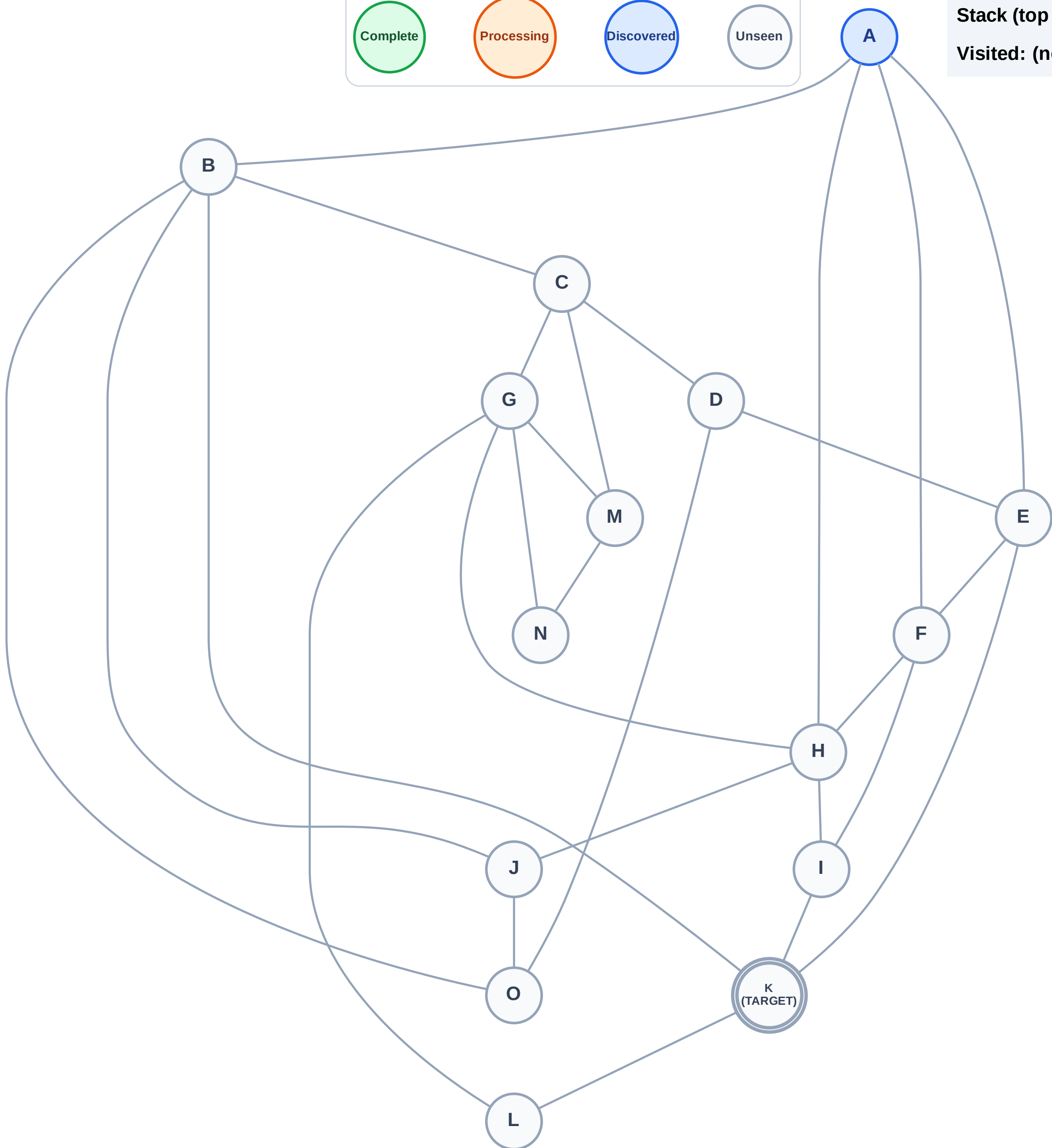
Step 1: Start at A (Seeking target node K)

Legend

The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

Unseen

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

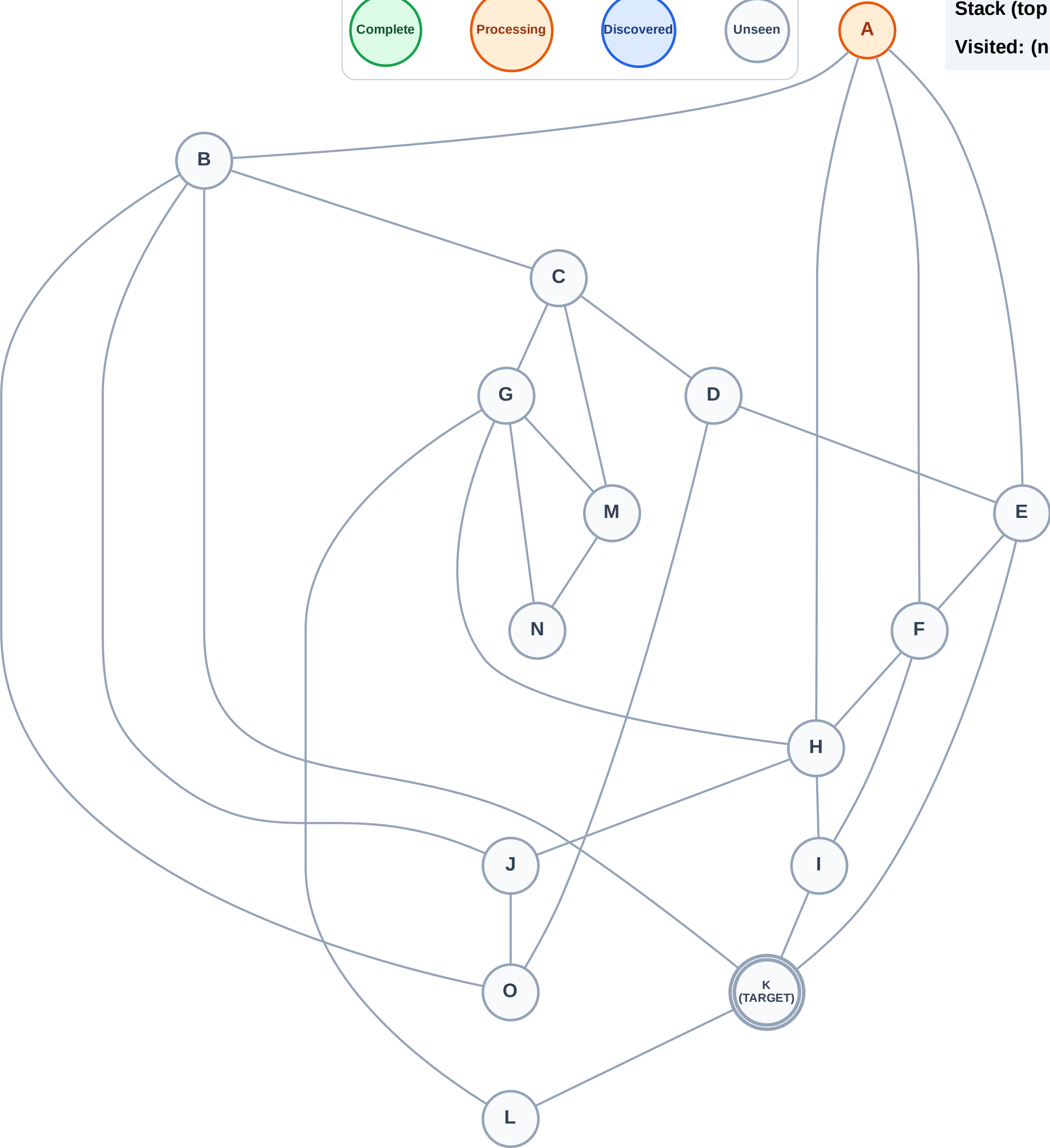
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



Step 3: Discover H, F, E, B from A

Step 3: Discover H, F, E, B from A

Legend

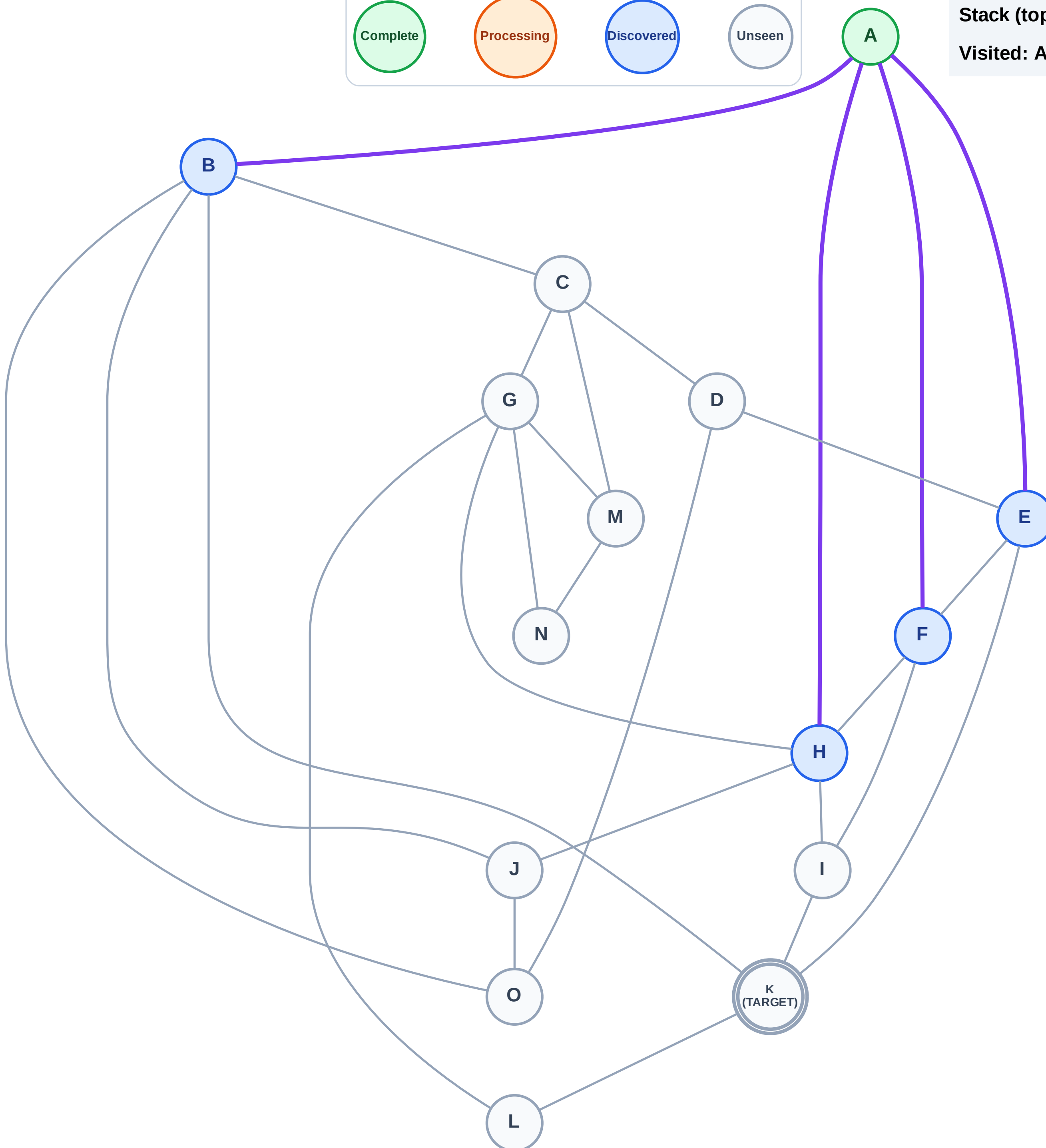
Complete Processing Discovered Unseen

Processing

Unseen

Unseen

Visited: A



Step 4: Process node B

Step 4: Process node B

Legend

Complete

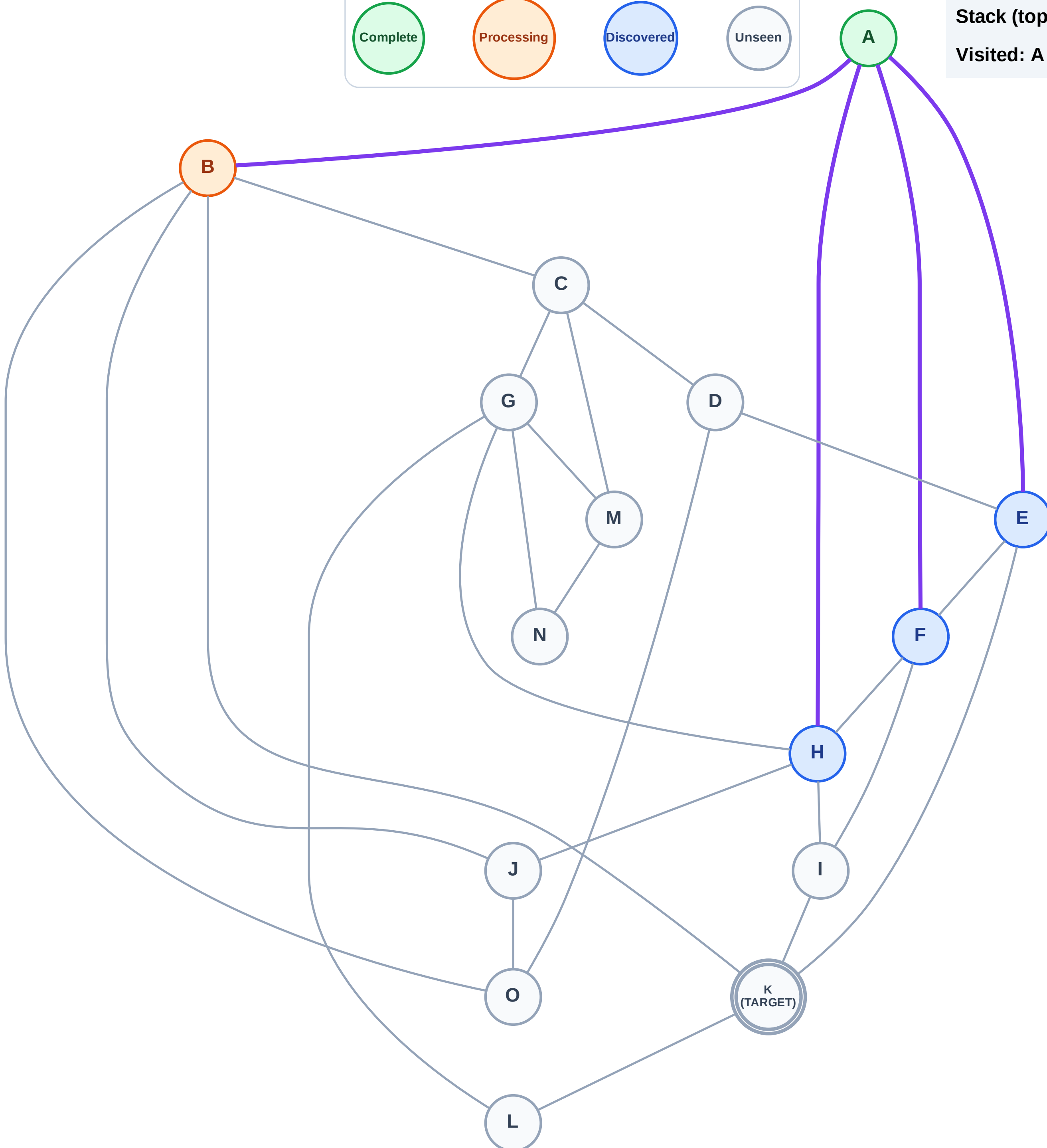
Processing

Discovered

Unseen

Stack (top at right): H | F | E

Visited: A



DFS Traversal

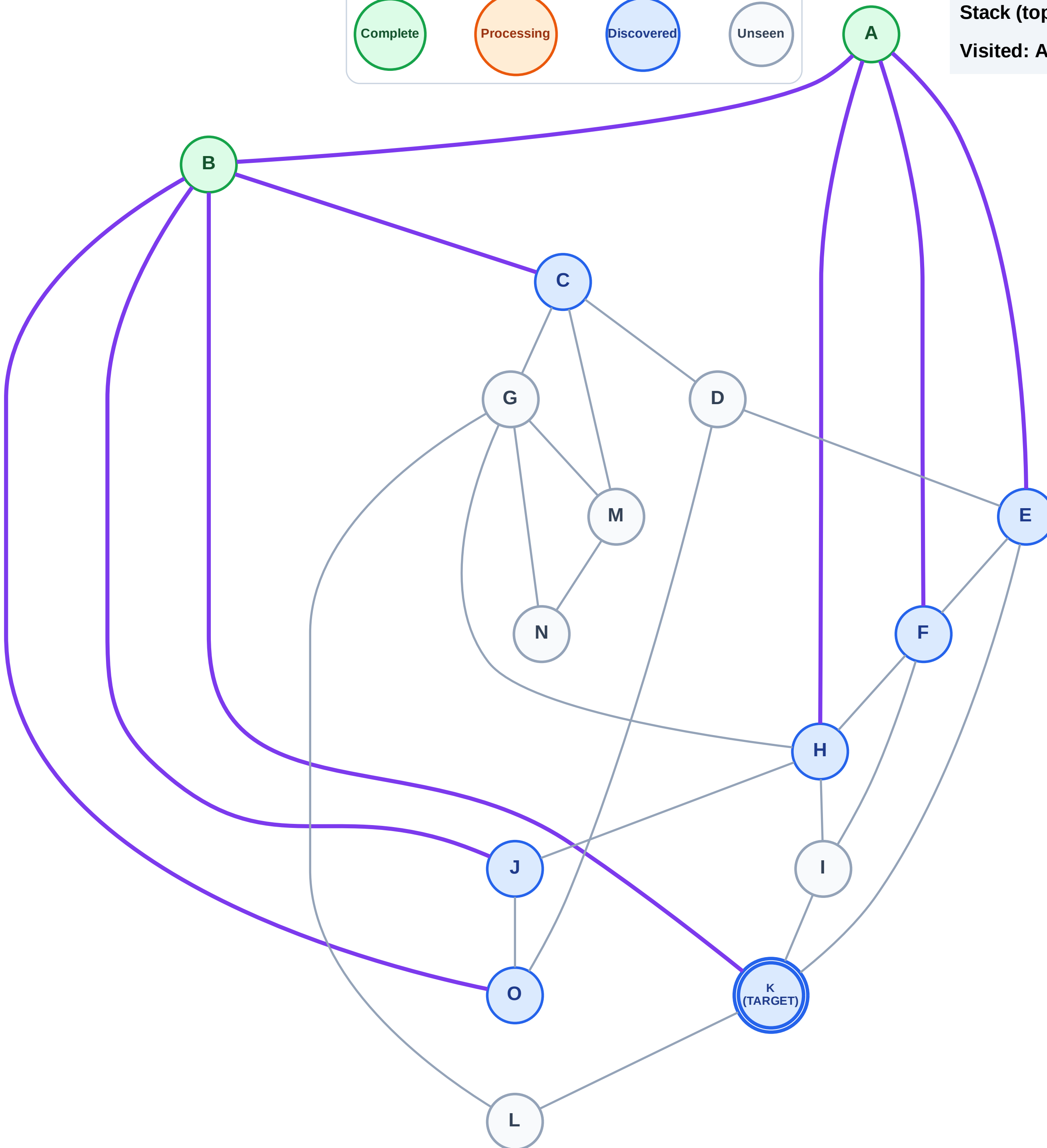
Step 5: Discover O, K, J, C from B

Legend



Stack (top at right): H | F | E | O | K | J | C

Visited: A, B



DFS Traversal

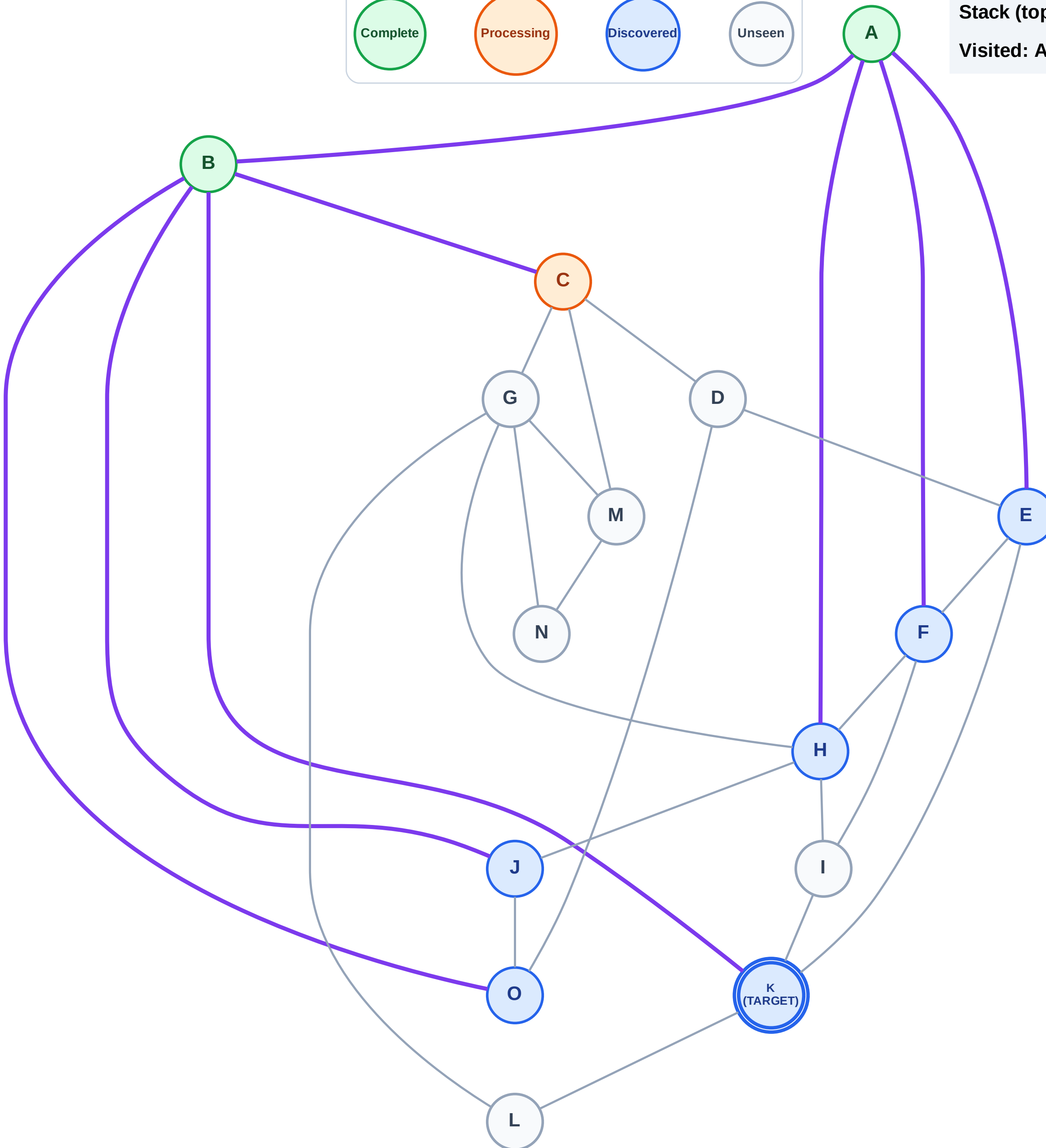
Step 6: Process node C

Legend



Stack (top at right): H | F | E | O | K | J

Visited: A, B



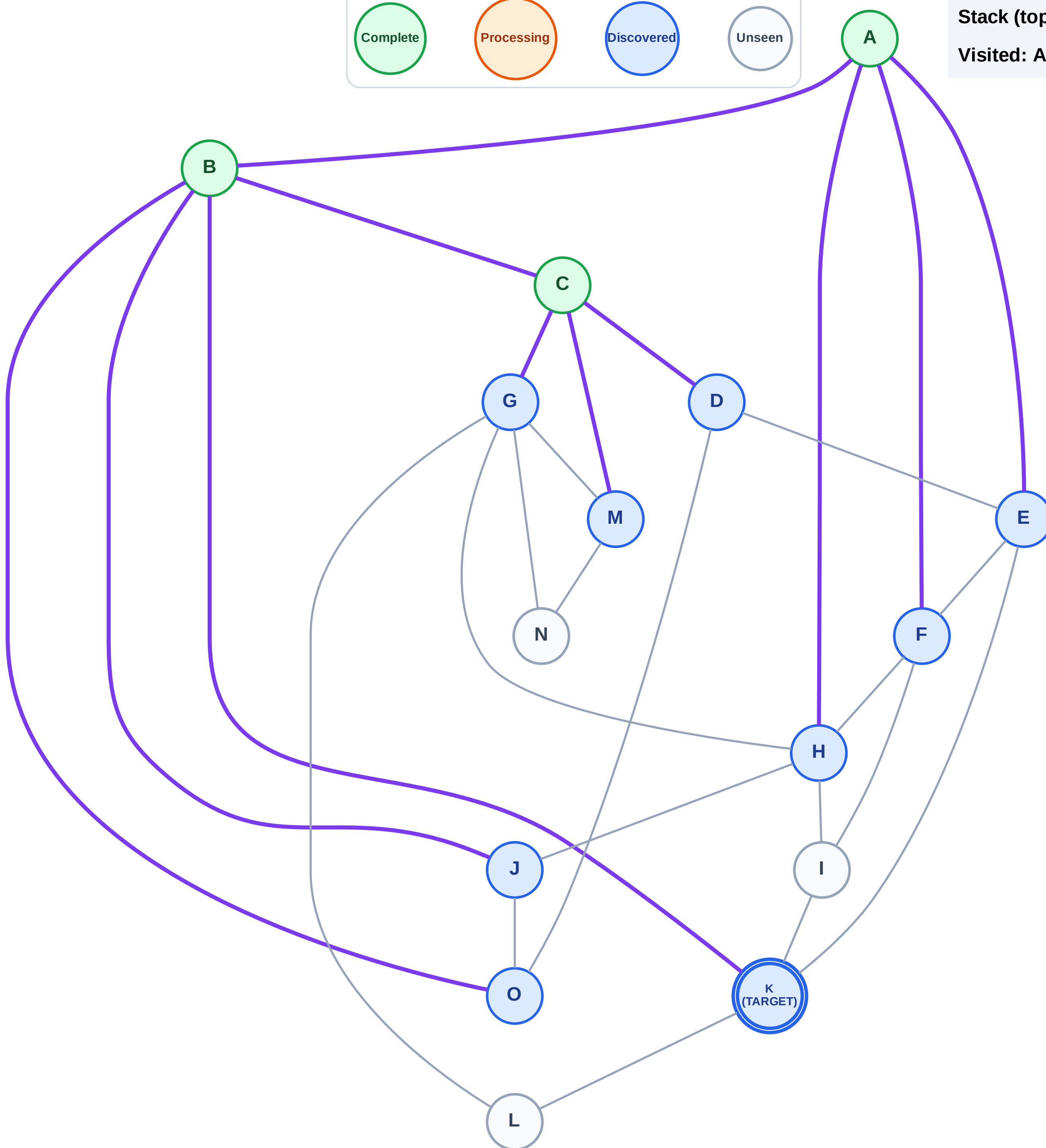
DFS Traversal

Step 7: Discover M, G, D from C

Legend



Stack (top at right): H | F | E | O | K | J | M | G | D
Visited: A, B, C



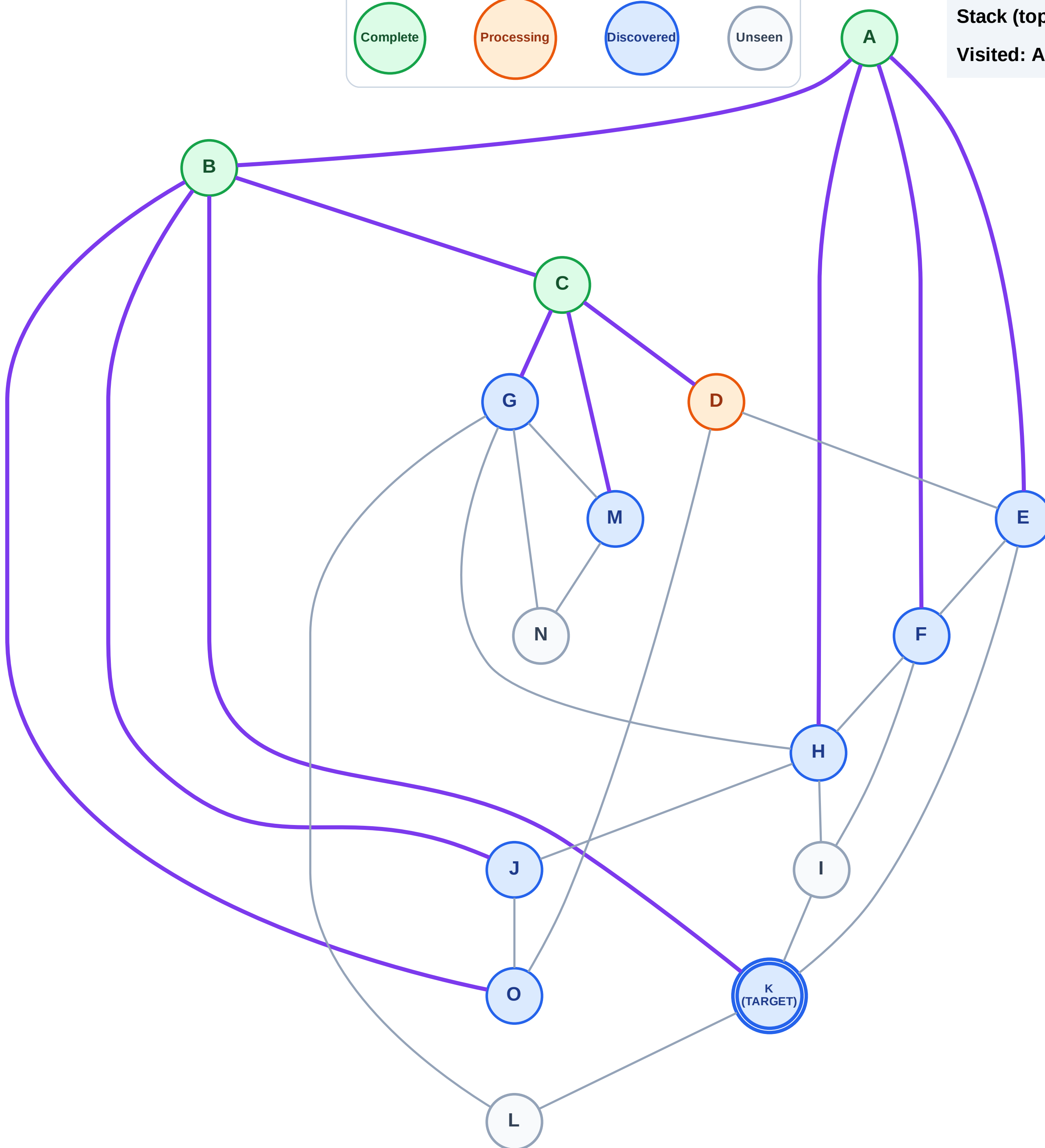
Step 8: Process node D

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Complete Processing Discovered Unseen

Unseen

Visited: A, B, C



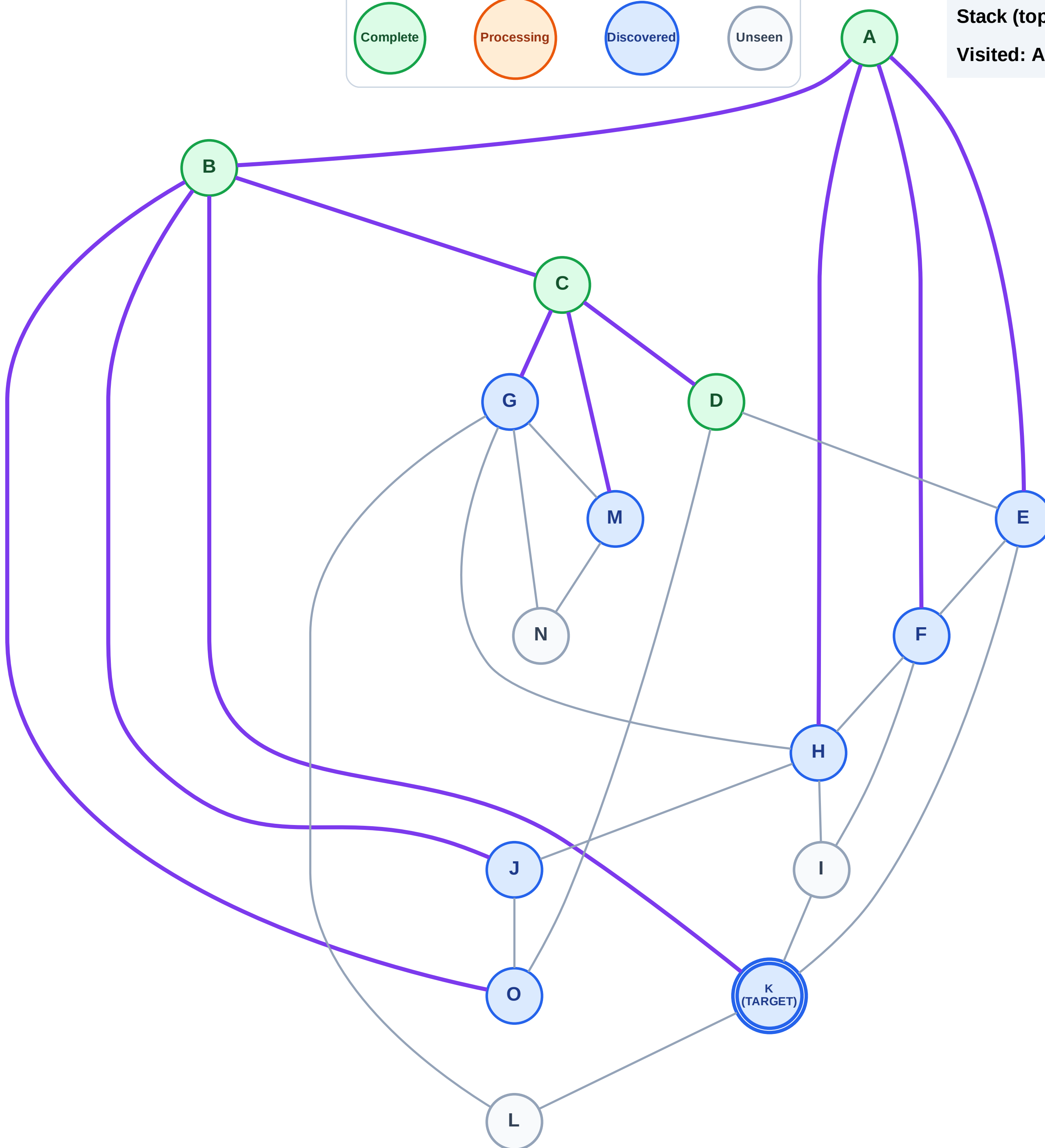
DFS Traversal

Step 9: No new neighbors from D

Legend



Stack (top at right): H | F | E | O | K | J | M | G
Visited: A, B, C, D



DFS Traversal

Step 10: Process node G

Legend



Complete



Processing



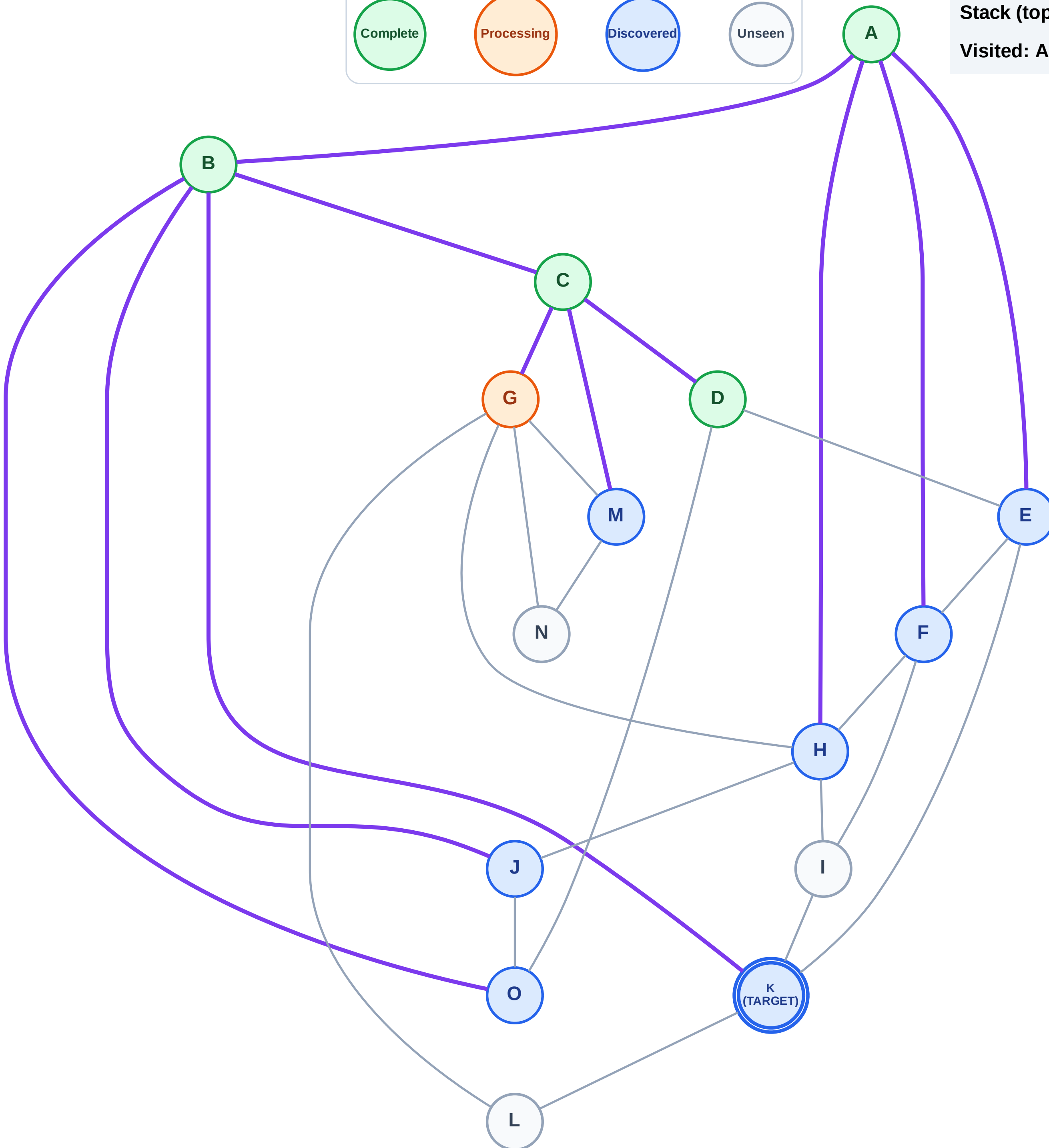
Discovered



Unseen

Stack (top at right): H | F | E | O | K | J | M

Visited: A, B, C, D



DFS Traversal

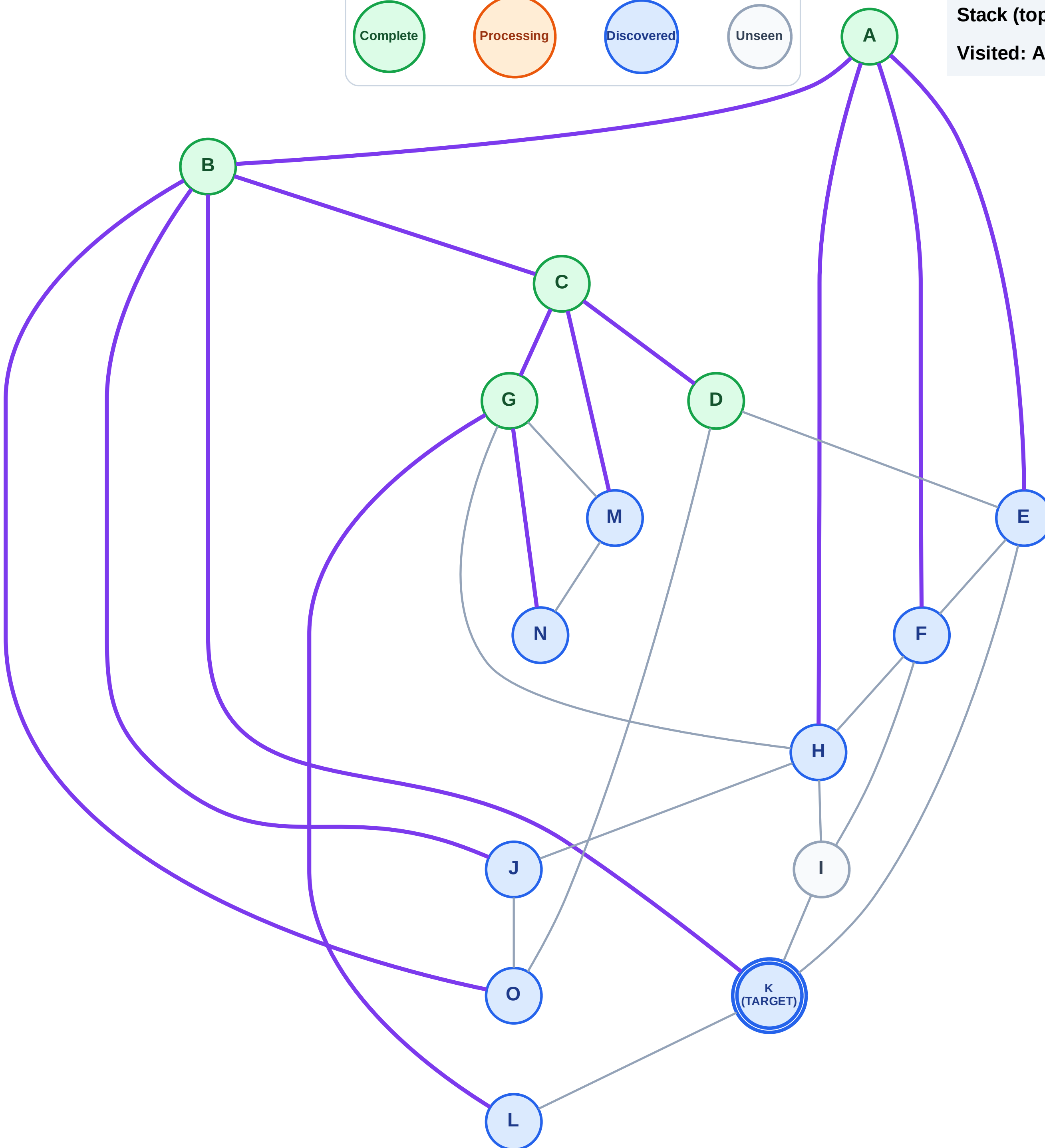
Step 11: Discover N, L from G

Legend



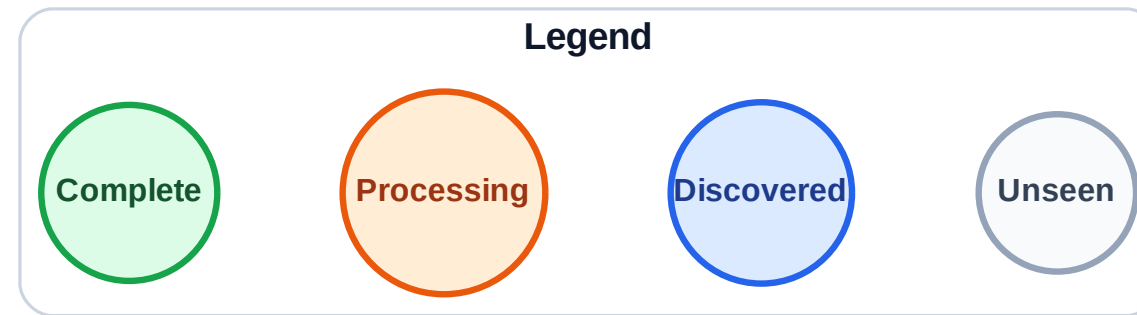
Stack (top at right): H | F | E | O | K | J | M | N | L

Visited: A, B, C, D, G

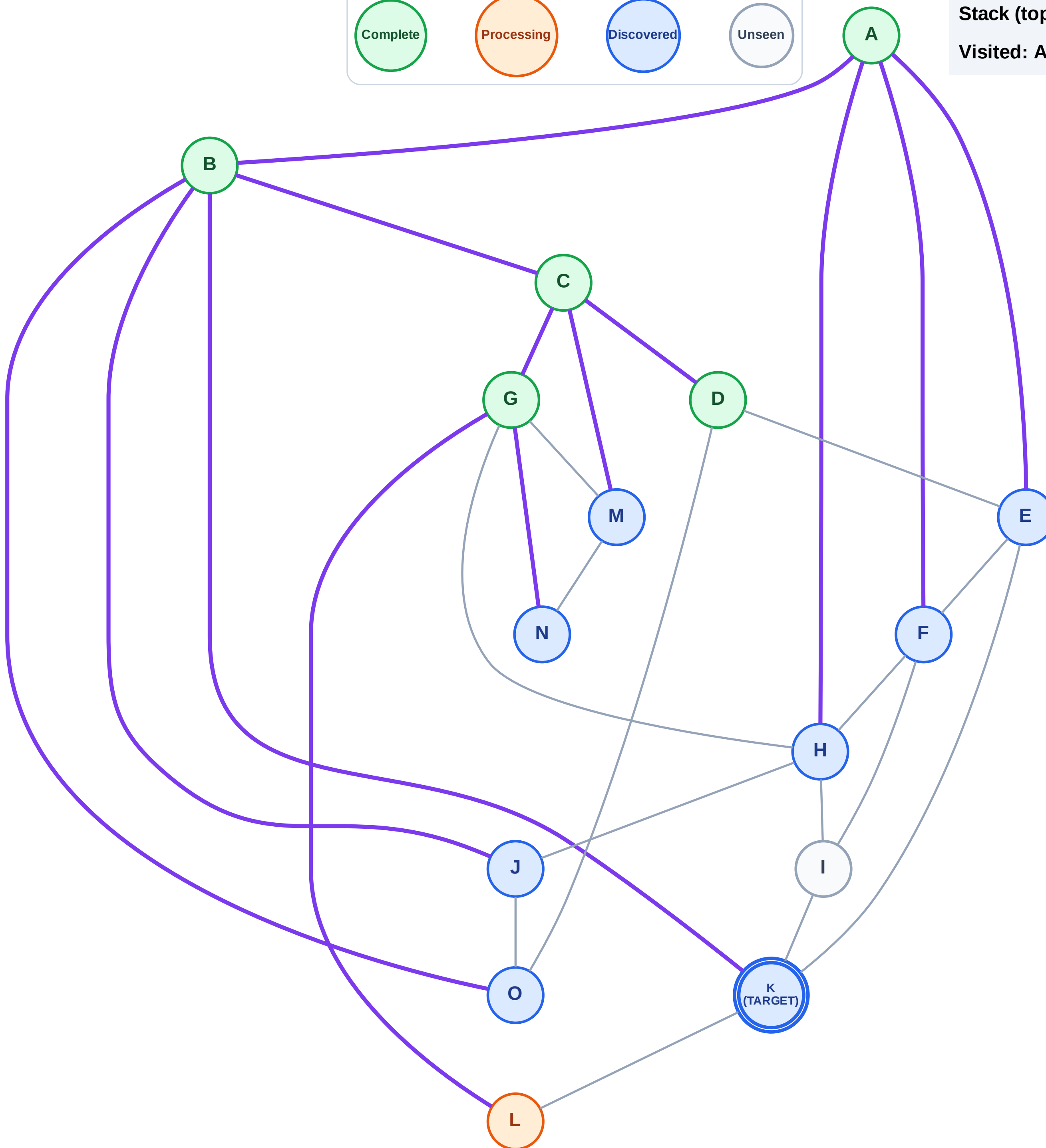


DFS Traversal

Step 12: Process node L



Stack (top at right): H | F | E | O | K | J | M | N
Visited: A, B, C, D, G



DFS Traversal

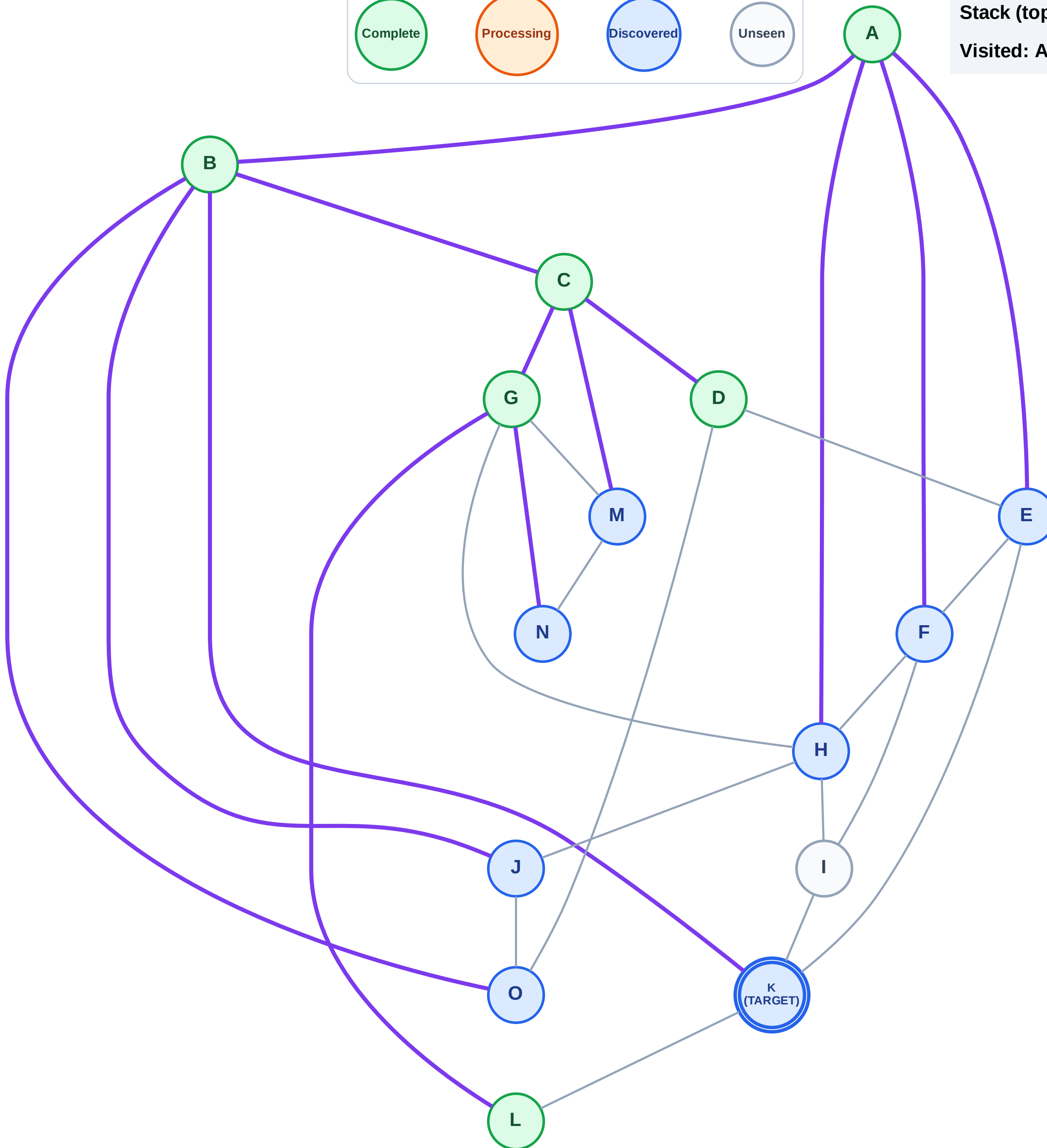
Step 13: No new neighbors from L

Legend



Stack (top at right): H | F | E | O | K | J | M | N

Visited: A, B, C, D, G, L



DFS Traversal

Step 14: Process node N

Legend



Complete



Processing



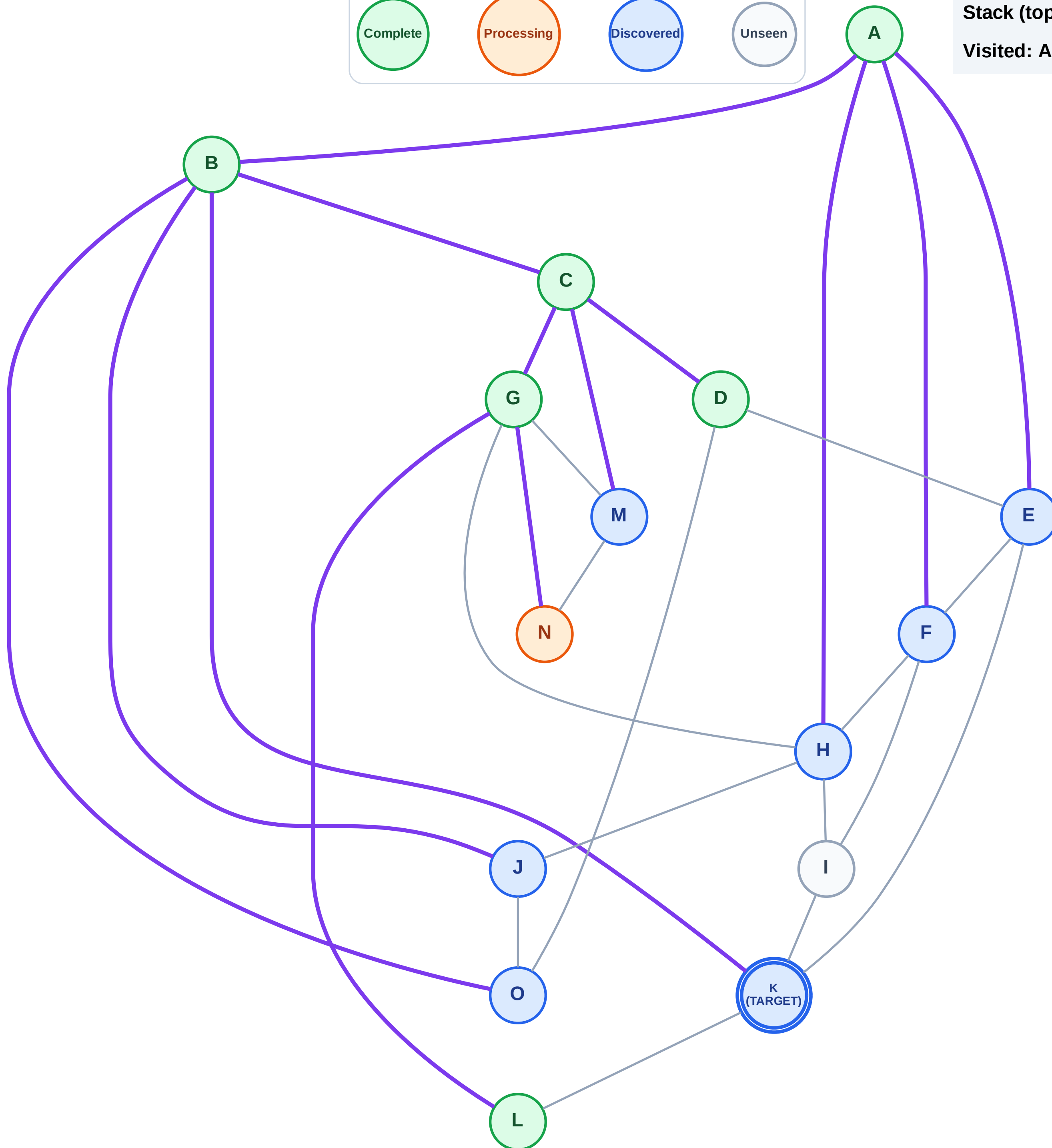
Discovered



Unseen

Stack (top at right): H | F | E | O | K | J | M

Visited: A, B, C, D, G, L



DFS Traversal

Step 15: No new neighbors from N

Legend



Complete



Processing



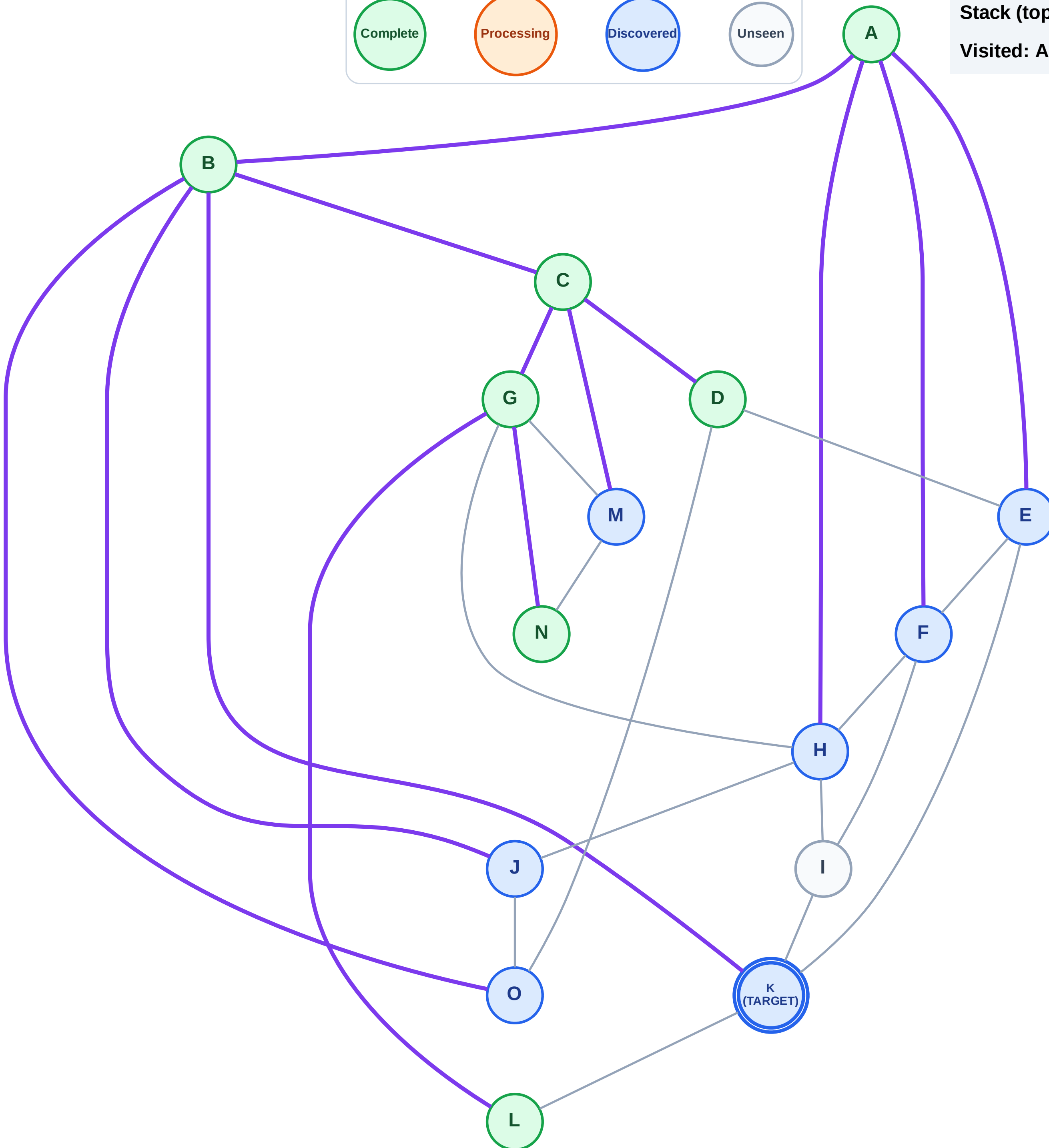
Discovered



Unseen

Stack (top at right): H | F | E | O | K | J | M

Visited: A, B, C, D, G, L, N



DFS Traversal

Step 16: Process node M

Legend



Complete



Processing



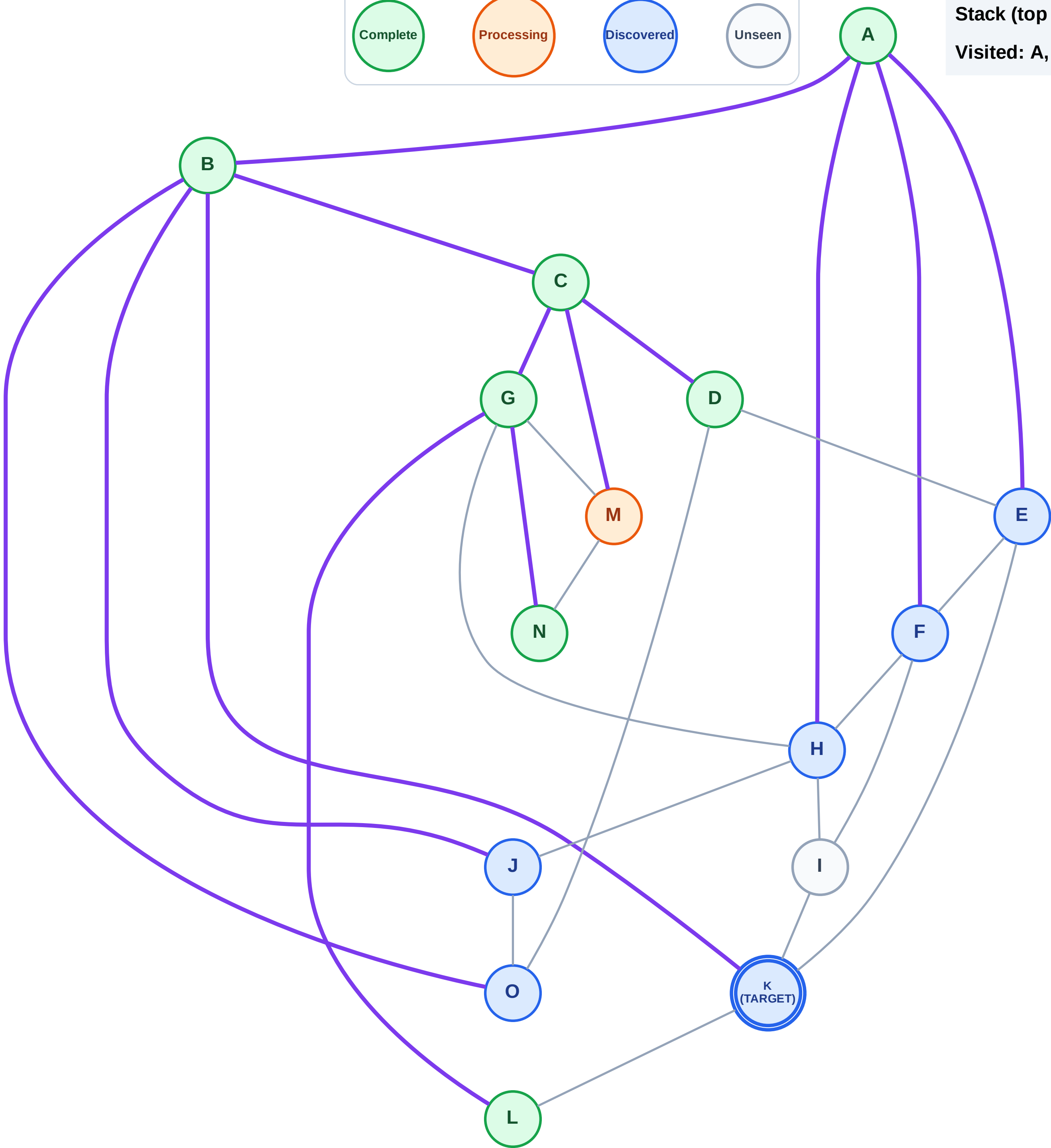
Discovered



Unseen

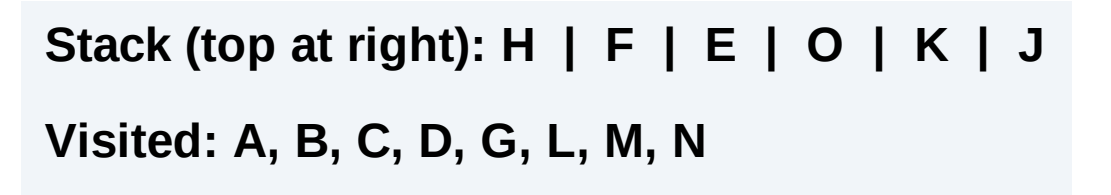
Stack (top at right): H | F | E | O | K | J

Visited: A, B, C, D, G, L, N



Step 17: No new neighbors from M

Legend



DFS Traversal

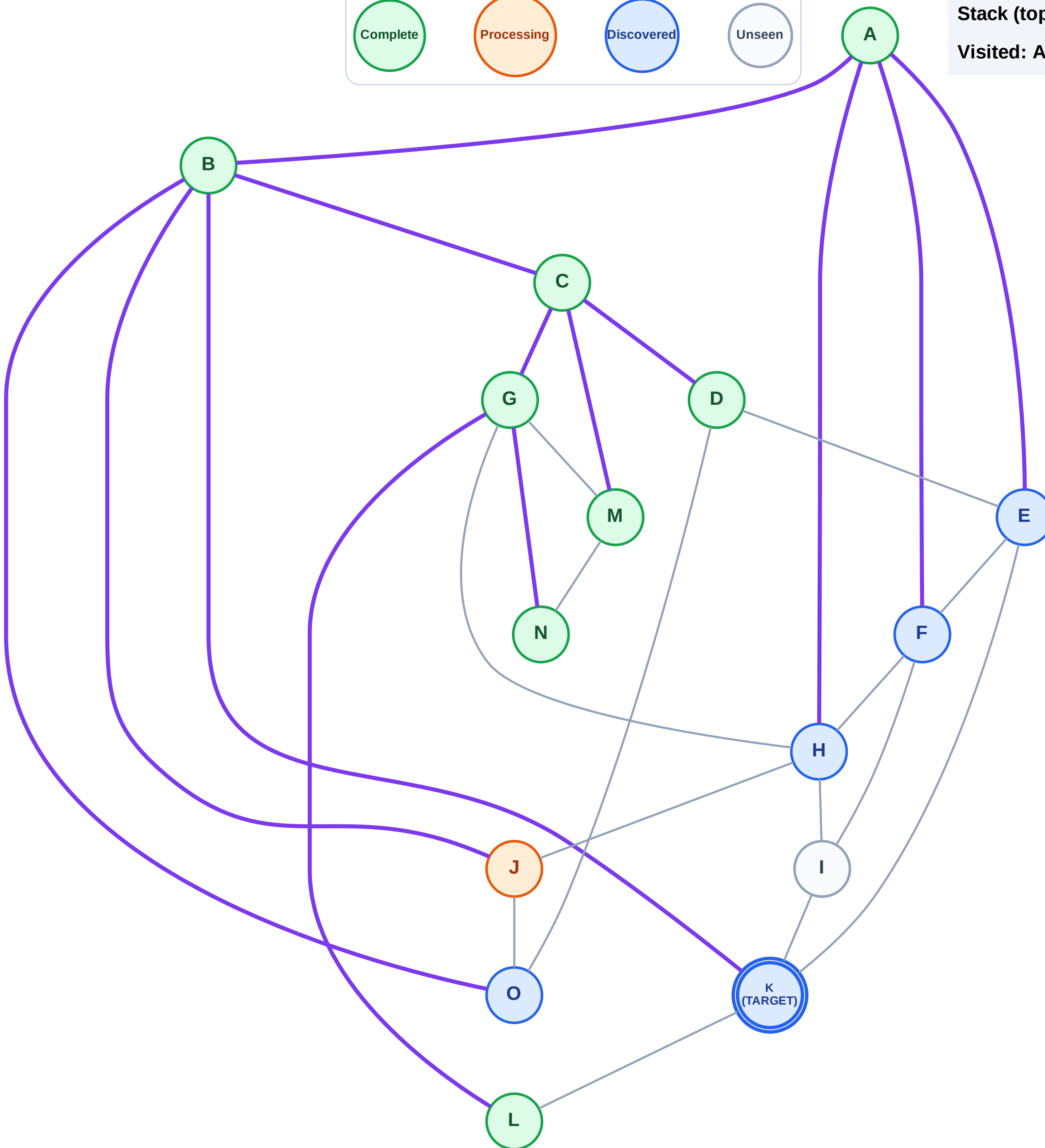
Step 18: Process node J

Legend



Stack (top at right): H | F | E | O | K

Visited: A, B, C, D, G, L, M, N



DFS Traversal

Step 19: No new neighbors from J

Legend



Complete



Processing



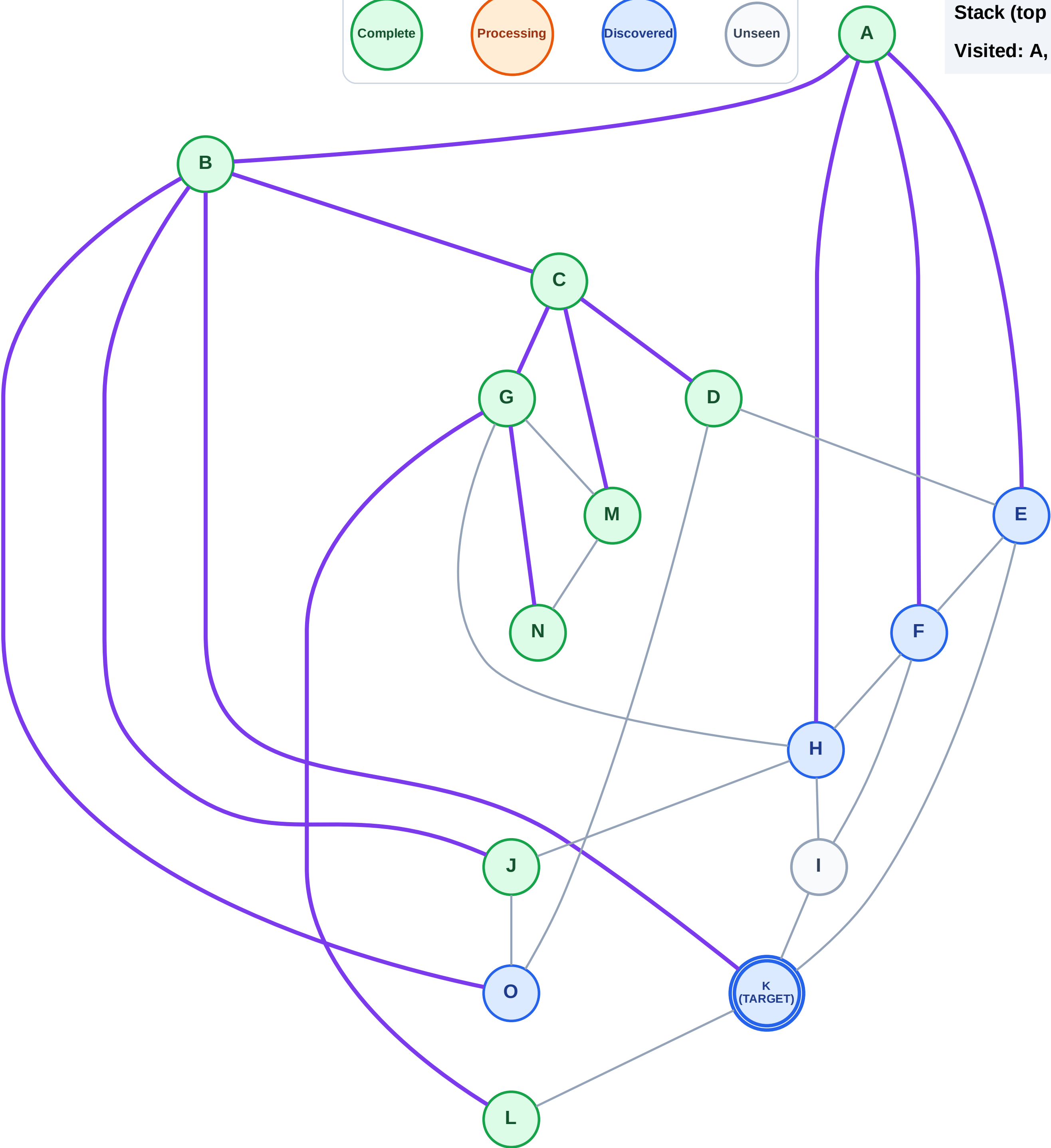
Discovered



Unseen

Stack (top at right): H | F | E | O | K

Visited: A, B, C, D, G, J, L, M, N



Step 20: Process node K

Legend

Complete



Processing



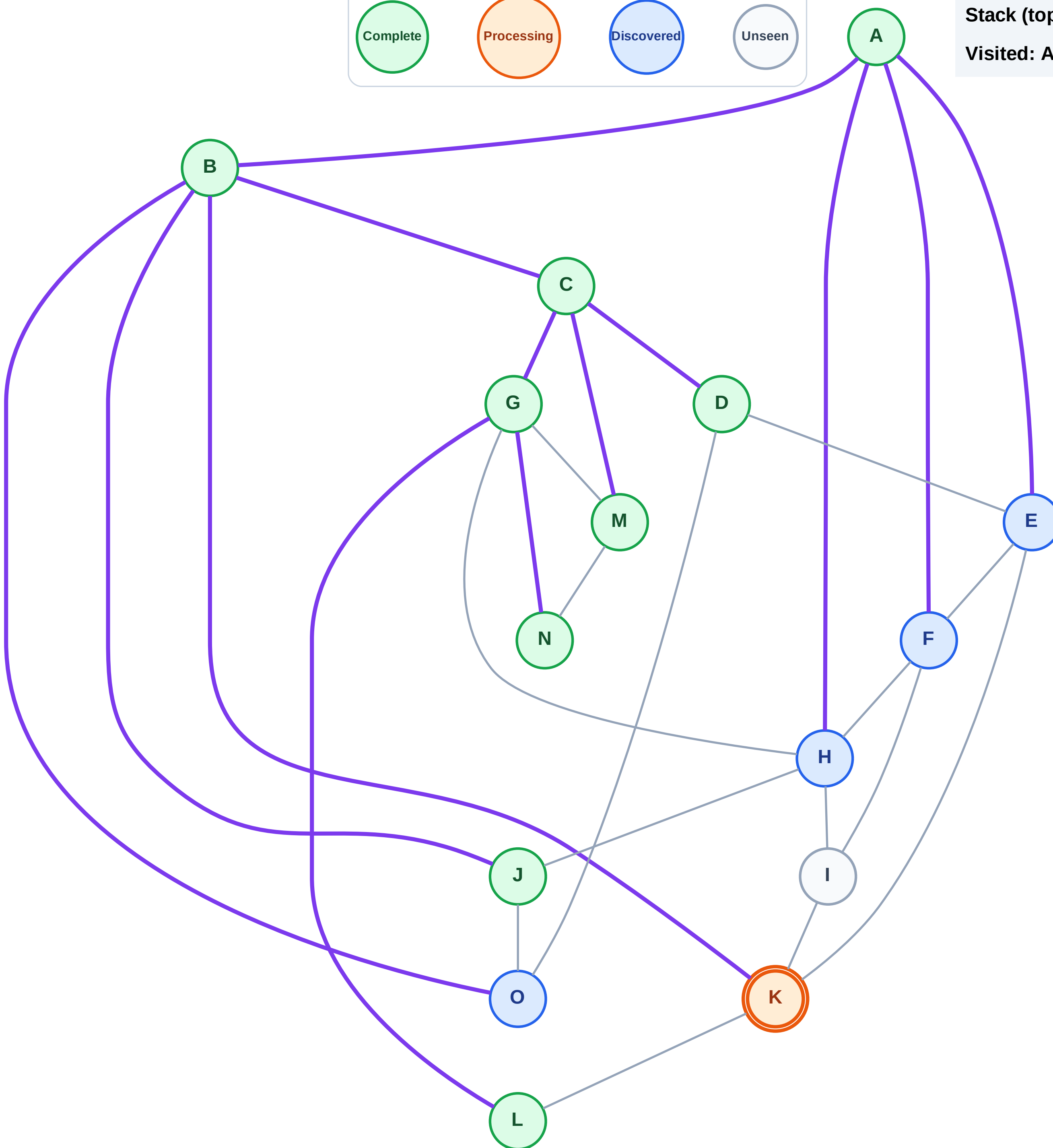
Discovered



Stack (top at right): H | F | E | O

Visited: A, B, C, D, G, J, L, M, N

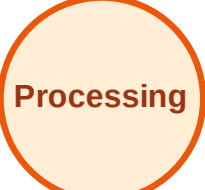
Visited: A, B, C, D, G, J, L, M, N



DFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Stack (top at right): H | F | E | O

Visited: A, B, C, D, G, J, K, L, M, N

